Chebyshev-Exact Acceleration under Hessian Variation, I: Sine–Jacobi Method

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Abstract

Knowing only a quadratic's spectral interval, the best worst-case error reduction by fixed-coefficient gradient methods is described by a Chebyshev polynomial, which many methods attain. They coincide on every fixed quadratic but respond differently when the Hessian changes between gradient evaluations, as along a nonquadratic trajectory or under link failures in distributed averaging. Let $\epsilon_N^\star$ be the ideal contraction on $[\mu, L]$. We measure the first-order response by its exact Lipschitz gain A_N. The normalized gain $(L-\mu)A_N/\epsilon_N^\star$ is at least of order N^2 without momentum; the classical Chebyshev semi-iterative method achieves order $N^{3/2}$ with constant about 2.31. Describing positive two-term methods with the same final polynomial by Jacobi matrices, we construct the sine–Jacobi method, with constant about 2.14, within seven percent of a universal lower bound. Beyond first order, both methods attain the optimal worst-case bounds of a widened spectral interval, yielding local linear convergence for objectives with Lipschitz Hessian. For a smooth nonquadratic energy and for gossip averaging with link failures the leading error term is explicit, and experiments confirm both the predicted errors, where the leading terms apply, and the predicted ordering of the methods beyond.

Keywords: Chebyshev acceleration ; Hessian variation ; orthogonal polynomials ; Jacobi matrices ; momentum methods ; strongly convex optimization

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1 Introduction

In a memoir published in 1854, Chebyshev, studying the steam-engine linkages of James Watt, asked which monic polynomial of given degree deviates least from zero on an interval [4]. The answer became the starting point of approximation theory [26], and a century later of iterative methods as well. When a quadratic function is minimized with one gradient per step, the error after N steps is a polynomial in the Hessian applied to the initial error, and if only an interval containing the spectrum is known, the best polynomial is a rescaled Chebyshev polynomial. This observation underlies the Chebyshev semi-iterative method of Golub and Varga [10, 11, 33, 27], Polyak's heavy-ball method as its stationary limit [25], and the benchmark for accelerated methods in optimization [21, 5].

The Chebyshev polynomial fixes *what* a method does after N steps on a fixed quadratic; it does not fix *how* the method gets there. Gradient descent with the reciprocals of the Chebyshev roots as stepsizes gives the same final iterate for every order of the steps, yet in floating point some orders amplify rounding errors catastrophically. Finding stable orders was a research problem of the 1950s and 1960s; Lebedev and Finogenov developed stable orderings in the early 1970s [15, 16]. These ideas reappeared in optimization as fractal stepsize schedules [1]. The three-term recurrence of Golub and Varga is another realization of the same polynomial, with momentum, and its own rounding behaviour was analyzed by Gutknecht and Röllin [12]. A polynomial thus has many realizations, and realizations that agree on the ideal problem may differ greatly on a perturbed one.

The question. Along the trajectory of a nonquadratic objective the Hessian changes from one gradient evaluation to the next, and in randomized or distributed iterations the curvature seen by each gradient fluctuates. We ask: *among gradient methods that produce the Chebyshev polynomial on every fixed quadratic, how do the responses to a Hessian that changes between gradient evaluations differ, and which method responds least?* To first order, a perturbation of the Hessian at step s that maps one eigenvector of the reference Hessian to another is multiplied by a scalar kernel, which depends on the two eigenvalues and on s and is invisible on a fixed quadratic. Its ℓ_2 norm over time, maximized over pairs of eigenvalues, is the *gain* A_N of the method: the exact Lipschitz constant of the linear response (Proposition 3.1).

Main results. We compare methods with the same optimal worst-case contraction factor $\epsilon_N^\star$ on fixed quadratics. For the two momentum methods below A_N is of order $N^{3/2}\epsilon_N^\star/(L-\mu)$, and the factor in front of this quantity is the *constant* of the method.

(1) Without momentum the gain is at least of order $N^2\epsilon_N^\star/(L-\mu)$ for every order of the Chebyshev stepsizes (Theorem 3.1).
(2) The semi-iterative method, the only prefix-exact two-term method, has constant $4/\sqrt{3}$ (Theorem 4.1).
(3) Positive two-term methods optimal at the final step correspond to weights on the Chebyshev nodes; every realization has $A_N \geq 2N^{3/2}\epsilon_N^\star/(L-\mu)$ (Proposition 5.1).
(4) The sine weights give the sine–Jacobi method (6) with constant $2.1379\ldots$ (Theorem 6.2).
(5) Beyond first order both momentum methods respond to drift δ as to a widening of the interval by δ, optimally (Theorems 7.1 to 7.3).
(6) For smooth energies and gossip with link failures the leading error term is explicit and confirmed by experiments (Sections 8 and 9).

Related work. Stable orderings of Chebyshev steps [15, 1] and rounding analysis [12] treat additive errors in one realization; we perturb the Hessian and compare realizations. Stepsize schedules [2] and performance estimation [8, 31] optimize the worst-case error over a function class; here the final polynomial is fixed. IQC analysis [17] provides a general framework for optimization dynamics. Inexact-oracle models [7] and noise-robust acceleration [3, 19] study gradient inaccuracies and additive noise; a Hessian perturbation here propagates with two spectral parameters. The sine weights lead to Rogers polynomials at a root of unity [14, 29].

All proofs are in the appendices.

2 Preliminaries

Notation. Horizons N are positive integers. X is a finite-dimensional real Hilbert space, $\mathrm{End}(X)$ the space of linear operators on it, $\|\cdot\|$ the operator norm and $\|E\|_{\mathrm{HS}} = (\operatorname{tr} E^\top E)^{1/2}$ the Hilbert–Schmidt norm. For a symmetric H, Π_λ is the orthogonal projector onto the eigenspace of $\lambda \in \mathrm{spec}(H)$, and $p(H) = \sum_\lambda p(\lambda)\Pi_\lambda$. We fix $0 < \mu < L$ and write $\Delta := L - \mu$, $\eta := (L+\mu)/\Delta > 1$, $\kappa := L/\mu$ and $z(\lambda) := (L+\mu-2\lambda)/\Delta$; the map z sends $[\mu, L]$ onto $[-1, 1]$, with $z(\mu) = 1$, $z(L) = -1$, $z(0) = \eta$.

2.1 Chebyshev polynomials

The Chebyshev polynomials $T_n(\cos\theta) = \cos n\theta$ and $U_n(\cos\theta) = \sin(n+1)\theta/\sin\theta$, extended continuously when $\sin\theta = 0$, both satisfy $y_{n+1} = 2xy_n - y_{n-1}$, with $T_0 = U_0 = 1$, $T_1 = x$, $U_1 = 2x$. We use [26, 18]: (T1) $|T_n| \leq 1 = T_n(1)$ and $|U_n| \leq n+1 = U_n(1)$ on $[-1, 1]$; (T2) for $n \geq 1$, the zeros of T_n are the nodes $\zeta_j = \cos\theta_j$, $\theta_j = (2j-1)\pi/(2n)$, and $T_n = 2^{n-1}\prod_j (x - \zeta_j)$; (T3) $T_n(\cosh a) = \cosh na$; (T4) $T_n' = nU_{n-1}$ and $T_n'(1) = n^2$. By Chebyshev's minimax theorem, $T_n(x)/T_n(\eta)$ has the least maximum modulus on $[-1, 1]$ among polynomials of degree at most n equal to 1 at $\eta > 1$ [26].

2.2 Gradient methods on quadratics

For $f_H(x) = \frac{1}{2}\langle H(x-x^\star), x-x^\star\rangle$ with $\operatorname{spec}(H) \subset [\mu, L]$, every affine-consistent fixed-coefficient realization (Definition 3.1) satisfies $x_N - x^\star = p_N(H)(x_0 - x^\star)$, where the *residual polynomial* p_N has degree at most N and $p_N(0) = 1$ [21]. By the minimax theorem the best worst-case contraction is attained by $p_N^\star(\lambda) = T_N(z(\lambda))/T_N(\eta)$, with $\max_{[\mu,L]} |p_N^\star| = \epsilon_N^\star := 1/T_N(\eta) \le 2q^N$, where $q = (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ by (T3). A method is *(final-)Chebyshev-exact* if $p_N = p_N^\star$. Three families of such methods appear in this paper. *Gradient descent* $x_{s+1} = x_s - \gamma_s \nabla f(x_s)$ has $p_N = \prod_s (1-\gamma_s\lambda)$, so it is Chebyshev-exact if and only if the γ_s^{-1} are the roots $\lambda_j = \frac{\Delta}{2}(\eta - \zeta_j)$ of $p_N^\star$ in some order. The *Chebyshev semi-iterative method* [10, 33, 27], with $\beta_t := T_{t-1}(\eta)/T_{t+1}(\eta)$,

$$x_1 = x_0 - \tfrac{2}{L+\mu}\nabla f(x_0), \qquad x_{t+1} = x_t + \beta_t(x_t - x_{t-1}) - (1+\beta_t)\tfrac{2}{L+\mu}\nabla f(x_t), \tag{1}$$

satisfies $x_t - x^\star = p_t^\star(H)(x_0 - x^\star)$ for *every* t; we call it *prefix-exact*. As $t \to \infty$ it tends to Polyak's heavy-ball method with optimal parameters. More generally, a *two-term* (momentum) method $x_{s+1} = x_s + \beta_s(x_s - x_{s-1}) - \gamma_s \nabla f(x_s)$, with $x_{-1} = x_0$ and $\beta_0 = 0$, has residual polynomials with $p_{s+1} = (1 + \beta_s - \gamma_s\lambda)p_s - \beta_s p_{s-1}$. Requiring $p_N = p_N^\star$ leaves the intermediate polynomials free; For positive steps and subsequent momentum, Section 5 identifies this freedom with a probability measure on the Chebyshev nodes.

2.3 Orthogonal polynomials and Jacobi matrices

A *Jacobi matrix* J of size N is real symmetric tridiagonal with diagonal $b_0, \dots, b_{N-1}$ and off-diagonal $a_1, \dots, a_{N-1} > 0$; rows and columns are indexed by $0, \dots, N-1$, and $J_{a:b}$ is the principal block on the sites $a, \dots, b$. The characteristic polynomials $P_s(x) = \det(xI - J_{0:s-1})$ of the leading blocks satisfy

$$P_{s+1}(x) = (x - b_s)P_s(x) - a_s^2 P_{s-1}(x), \qquad P_{-1} = 0,\ P_0 = 1, \tag{2}$$

and $P_N(x) = \det(xI - J)$ [30, 32]. For instance, $b_s = 0$, $a_1^2 = \frac{1}{2}$, $a_s^2 = \frac{1}{4}$ $(s \ge 2)$ give $P_s = 2^{1-s}T_s$ for $s \ge 1$. If J has eigenvalues ζ_j and normalized eigenvectors v_j, the weights $w_j := (v_j)_0^2$ are positive, sum to 1, and $P_0, \dots, P_{N-1}$ are orthogonal with respect to $\sum_j w_j \delta_{\zeta_j}$. Conversely (finite Favard theorem), every probability measure on N distinct real points arises in this way from exactly one Jacobi matrix [32]. Hence the Jacobi matrices with spectrum $\{\zeta_1, \dots, \zeta_N\}$ correspond one-to-one to the probability vectors w with $w_j > 0$. For the Chebyshev example the spectrum consists of the nodes (T2) and $w_j = 1/N$ (Gauss–Chebyshev quadrature [18]). Deleting row and column s of $xI - J$ leaves a block-diagonal matrix, so with $Q_{N-s-1}^{(s+1)}(x) := \det(xI - J_{s+1:N-1})$, $Q_0^{(N)} := 1$,

$$\big[(xI - J)^{-1}\big]_{ss} = P_s(x)\, Q_{N-s-1}^{(s+1)}(x)/P_N(x), \qquad x \notin \operatorname{spec}(J): \tag{3}$$

the diagonal of the resolvent (the *Green function*) is a product of a left and a right characteristic polynomial. Finally, with the reversal $Re_s = e_{N-1-s}$, J is *persymmetric* if $RJR = J$; a Jacobi matrix is persymmetric if and only if its weights are $w_j = c/|P_N'(\zeta_j)|$ with a constant $c > 0$, and it is then determined by its spectrum [6]. All leading polynomials of a Jacobi matrix with $\operatorname{spec}(J) \subset [-1, 1]$ are positive on $(1, \infty)$, because their zeros interlace those of P_N [30].

3 Gradient Methods, Hessian Drift, and the Gain A_N

3.1 A general model of a gradient method

The following state-space format [17] describes affine-consistent methods with one gradient call per step and fixed linear combinations of m stored vectors.

Definition 3.1 (Realization)**.** A *realization* of length N with memory m is a tuple $\mathcal{R} = (m, b, \ell, \{M_s, g_s, q_s\}_{s=0}^{N-1})$ with $b, \ell, g_s, q_s \in \mathbb{R}^m$, $M_s \in \mathbb{R}^{m\times m}$ and $\ell^\top b = 1$. Put $c_s = M_{s-1}\cdots M_0 b$,

with $c_0 = b$. We require affine consistency: $q_s^\top c_s = 1$ and $\ell^\top c_N = 1$. It acts on states $Y_s \in \mathbb{R}^m \otimes X$, i.e. on m-tuples of vectors of X, by $Y_0 = b \otimes x_0$, $x_N = (\ell^\top \otimes I_X)Y_N$ and

$$Y_{s+1} = (M_s \otimes I_X)Y_s - (g_s \otimes I_X)\,\nabla f\big((q_s^\top \otimes I_X)Y_s\big).$$

For example, the two-term method $x_{s+1} = x_s + \beta_s(x_s - x_{s-1}) - \gamma_s \nabla f(x_s)$ with $Y_s = (x_s, x_{s-1})$, $x_{-1} := x_0$, has $m = 2$, $b = (1,1)^\top$, $\ell = q_s = (1,0)^\top$, $g_s = (\gamma_s, 0)^\top$ and $M_s = \left(\begin{smallmatrix} 1+\beta_s & -\beta_s \\ 1 & 0 \end{smallmatrix}\right)$; gradient descent has $m = 1$, $b = \ell = q_s = M_s = 1$, $g_s = \gamma_s$.

On a quadratic, put $\widehat{Y}_s := Y_s - c_s \otimes x^\star$. If $Hv = \lambda v$ and $\widehat{Y}_s = y \otimes v$, affine consistency gives $\widehat{Y}_{s+1} = (C_s(\lambda)y) \otimes v$ with the *transfer matrix* $C_s(\lambda) := M_s - \lambda g_s q_s^\top$. Consequently, for $f = f_H$, the error satisfies $x_N - x^\star = p_{\mathcal{R},N}(H)(x_0 - x^\star)$ with the residual polynomial $p_{\mathcal{R},N}(\lambda) = \ell^\top C_{N-1}(\lambda) \cdots C_0(\lambda)\, b$. Affine consistency gives $p_{\mathcal{R},N}(0) = 1$. The realization is *final-Chebyshev-exact* if $p_{\mathcal{R},N} = p_N^\star$.

3.2 Hessian drift and the first-order kernel

Now suppose that the gradient queried at step s is that of a quadratic with Hessian $H_s = H + E_s$ instead of H, where $\mu I \preceq H \preceq LI$ is a fixed reference Hessian. Then the error is mapped by the ordered product

$$\mathcal{T}_{\mathcal{R}}(H_0, \dots, H_{N-1}) = (\ell^\top \otimes I_X)\,\mathsf{C}_{N-1}(H_{N-1}) \cdots \mathsf{C}_0(H_0)\,(b \otimes I_X), \tag{4}$$

where $\mathsf{C}_s(A) := M_s \otimes I_X - g_s q_s^\top \otimes A$, so that $x_N - x^\star = \mathcal{T}_{\mathcal{R}}(H_0, \dots, H_{N-1})(x_0 - x^\star)$. For a sequence $E = (E_0, \dots, E_{N-1})$ of operators we write $\mathcal{T}_{\mathcal{R}}(H + E)$ for $\mathcal{T}_{\mathcal{R}}(H + E_0, \dots, H + E_{N-1})$. Without drift, $\mathcal{T}_{\mathcal{R}}(H, \dots, H) = p_{\mathcal{R},N}(H)$. With $E_s = \delta B_s$, the first variation $\mathcal{T}'_{\mathcal{R},H}[B] = \frac{d}{d\delta}\mathcal{T}_{\mathcal{R}}(H + \delta B)|_{\delta=0}$ is a sum over s of terms $\mathcal{T}'_s(B_s)$ in which the s-th factor is replaced by $-g_s q_s^\top \otimes B_s$. On the block $\Pi_\lambda B_s \Pi_\nu$, which maps the ν-eigenspace of H to the λ-eigenspace, the factors before step s act with eigenvalue ν and those after it with eigenvalue λ.

Definition 3.2 (First-order kernel)**.** The *first-order kernel* of $\mathcal{R}$ at step s is the function of $\lambda, \nu \in [\mu, L]$

$$K_{s,\mathcal{R}}(\lambda, \nu) := -\ell^\top C_{N-1}(\lambda) \cdots C_{s+1}(\lambda)\, g_s q_s^\top\, C_{s-1}(\nu) \cdots C_0(\nu)\, b,$$

with empty products equal to the identity.

The kernel separates a realization from its residual polynomial. On the diagonal its sum over time is determined by the polynomial: a uniform shift $B_s = I$ for all s gives $\sum_s K_{s,\mathcal{R}}(\lambda, \lambda) = p'_{\mathcal{R},N}(\lambda)$. Off the diagonal, and even on the diagonal for perturbations that are not uniform in time, it depends on the realization.

Lemma 3.1 (First-order expansion)**.** *For every realization, $\mathcal{T}'_{\mathcal{R},H}[B] = \sum_s \mathcal{T}'_s(B_s)$, and $\Pi_\lambda \mathcal{T}'_s(B_s)\Pi_\nu = K_{s,\mathcal{R}}(\lambda, \nu)\,\Pi_\lambda B_s \Pi_\nu$ for all $\lambda, \nu \in \operatorname{spec}(H)$.*

The proof is in Appendix A. Thus, block by block, the first-order response is the scalar sequence $(K_{s,\mathcal{R}}(\lambda, \nu))_s$ paired with the sequence of perturbation blocks. The natural size of this pairing is the ℓ_2 norm over time, and the natural size of the realization is its worst case over spectral pairs: the *gain* of $\mathcal{R}$ is $A_N(\mathcal{R}) := \sup_{\lambda,\nu \in [\mu,L]} \Big(\sum_{s=0}^{N-1} K_{s,\mathcal{R}}(\lambda, \nu)^2\Big)^{1/2}$, a property of the method, uniform over all Hessians with spectrum in $[\mu, L]$. The next proposition shows that A_N is the exact constant of the linear response, for operator-valued (not only scalar) perturbation blocks, and that the full response is a finite polynomial in the perturbations.

Proposition 3.1 (Finite drift and the size of the linear response)**.** *Let $\mu I \preceq H \preceq LI$ and let $E_0, \dots, E_{N-1} \in \operatorname{End}(X)$ be arbitrary. Then $\mathcal{T}_{\mathcal{R}}(H + E) = p_{\mathcal{R},N}(H) + \mathcal{T}'_{\mathcal{R},H}[E] + \mathcal{T}^{(\geq 2)}_{\mathcal{R},H}[E]$, where $\mathcal{T}^{(\geq 2)}_{\mathcal{R},H}$ is a sum of k-linear expressions in $(E_0, \dots, E_{N-1})$ with $2 \leq k \leq N$. Moreover $\big\|\mathcal{T}'_{\mathcal{R},H}[E]\big\|_{\mathrm{HS}} \leq A_N(\mathcal{R})\Big(\sum_{s=0}^{N-1} \|E_s\|_{\mathrm{HS}}^2\Big)^{1/2}$, and the constant $A_N(\mathcal{R})$ cannot be decreased: equality holds for $X = \mathbb{R}^2$, $H = \operatorname{diag}(\lambda_*, \nu_*)$ and $E_s = K_{s,\mathcal{R}}(\lambda_*, \nu_*)\, e_1 e_2^\top$, where (λ_*, ν_*) attains the supremum defining $A_N(\mathcal{R})$.*

The proof (Appendix A) is a Cauchy–Schwarz inequality on each block $\Pi_\lambda(\cdot)\Pi_\nu$ followed by summation over the blocks, which are orthogonal in the Hilbert–Schmidt inner product. For the two-term methods studied below the kernel factorizes (Theorem 5.2): $K_s(\lambda,\nu) = c_s\,\phi_s(\lambda)\,\psi_s(\nu)$ and $\mathcal{T}'_s(E) = c_s\,\phi_s(H)\,E\,\psi_s(H)$. If all $|\phi_s|$, and all $|\psi_s|$, attain their maxima at a common point, as for both momentum methods below, the bound holds in the operator norm: $\|\mathcal{T}'_{\mathcal{R},H}[E]\| \le \sum_s |c_s| \max|\phi_s| \max|\psi_s| \|E_s\| \le A_N(\mathcal{R})(\sum_s \|E_s\|^2)^{1/2}$.

3.3 Stepsize schedules

The simplest Chebyshev-exact methods have no momentum: gradient descent with the Chebyshev stepsizes $\gamma_s = \lambda_{j(s)}^{-1}$ in an order $j(\cdot)$. The order does not affect the residual polynomial, and the fractal order [1, 15] is the classical stable choice. No order makes the gain small.

Theorem 3.1 (Drift gain of exact stepsize schedules)**.** *For every $N \ge 1$ and every ordering of the Chebyshev stepsizes, $A_N \ge 2\sqrt{2/3}\,N^2\epsilon_N^\star/\Delta$.*

The proof (Appendix A) evaluates the kernel at $\lambda = \nu = \mu$, where its temporal ℓ_2 norm is independent of the order; with momentum the growth improves to $N^{3/2}$ (Section 4).

4 The Chebyshev Semi-Iterative Method

We now compute the kernel of the classical method (1). Its prefix exactness is classical [10, 33]; it is also the case $P_0 = 1$, $P_s = 2^{1-s}T_s$ for $s \ge 1$ of Theorem 5.1 below (Corollary 7.1). The kernel is the product of the residual of the ν-mode up to step s, the gradient coefficient at step s, and the response of the remaining steps to a unit impulse in the λ-mode, which is a Chebyshev polynomial of the second kind.

Theorem 4.1 (Kernel and gain of the semi-iterative method)**.** *For the method* (1)*, $K_0^{\mathrm{pref}}(\lambda,\nu) = -\frac{2\epsilon_N^\star}{\Delta}U_{N-1}(z(\lambda))$ and $K_s^{\mathrm{pref}}(\lambda,\nu) = -\frac{4\epsilon_N^\star}{\Delta}U_{N-s-1}(z(\lambda))T_s(z(\nu))$, $1 \le s \le N-1$. The supremum defining the gain is attained at $\lambda = \nu = \mu$, and $A_N^{\mathrm{pref}} \sim \frac{4}{\sqrt{3}}\frac{N^{3/2}}{\Delta}\epsilon_N^\star$.*

The factor $T_s(z(\nu))$, at most 1 in modulus, is the residual already achieved before the perturbation; the factor $U_{N-s-1}(z(\lambda))$, up to $N-s$, is the amplification of a fresh impulse by the remaining steps. Early perturbations are therefore the most harmful, and the worst pair is the slowest mode $\lambda = \nu = \mu$. Within the two-term class this gain is forced: a two-term method $x_{t+1} = \alpha_t x_t - \beta_t x_{t-1} - \gamma_t \nabla f(x_t)$ with residual polynomials $p_t^\star$ for all $t \le N$ has the coefficients of (1) (Appendix B). To lower the gain one must give up exactness at intermediate steps and keep it only at the final step.

5 Positive Two-Term Methods with the Chebyshev Terminal Polynomial

5.1 Jacobi coordinates

A two-term method with positive steps and subsequent momentum has residual polynomials satisfying a three-term recurrence. After the change of variable $x = z(\lambda)$ and a normalization, this recurrence becomes the monic recurrence (2) of a Jacobi matrix. Conversely, every Jacobi matrix whose characteristic polynomial has its zeros in $[-1,1]$ defines a two-term method:

Theorem 5.1 (Jacobi method)**.** *Let J be an $N \times N$ Jacobi matrix with* $\mathrm{spec}(J) \subset [-1,1]$*, and let P_s be its leading characteristic polynomials* (2)*, with $a_0 := 0$. Then $P_s(\eta) > 0$ for $0 \le s \le N$, and the method*

$$x_{s+1} = \tfrac{P_s(\eta)}{P_{s+1}(\eta)}\Big[(\eta - b_s)x_s - \tfrac{2}{\Delta}\nabla f(x_s)\Big] - a_s^2\,\tfrac{P_{s-1}(\eta)}{P_{s+1}(\eta)}\,x_{s-1}, \qquad 0 \le s \le N-1,$$

has residual polynomials $r_s(\lambda) = P_s(z(\lambda))/P_s(\eta)$ on every quadratic with $\mu I \preceq H \preceq LI$. In particular, it is final-Chebyshev-exact if and only if $P_N = 2^{1-N}T_N$, i.e. $\operatorname{spec}(J)$ *consists of the Chebyshev nodes $\zeta_1, \dots, \zeta_N$.*

Interlacing gives $P_s(\eta) > 0$ (Section 2.3). Conversely, every two-term method $x_{s+1} = \alpha_s x_s - \beta_s x_{s-1} - \gamma_s \nabla f(x_s)$ with $\alpha_s - \beta_s = 1$, $\gamma_s > 0$, $\beta_0 = 0$ and $\beta_s > 0$ for $s \geq 1$ has Jacobi coordinates: the monic polynomials $P_s := r_s \prod_{t<s} \frac{2}{\Delta\gamma_t}$, in $x = z(\lambda)$, satisfy (2) with $b_s = \eta - \frac{2\alpha_s}{\Delta\gamma_s}$ and $a_s^2 = \frac{4\beta_s}{\Delta^2\gamma_s\gamma_{s-1}} > 0$. By the correspondence of Section 2.3, the final-Chebyshev-exact two-term methods correspond one-to-one to the Jacobi matrices with $\det(xI - J) = 2^{1-N}T_N(x)$, and hence to the probability weights $w = (w_1, \dots, w_N)$, $w_j > 0$, on the Chebyshev nodes. We call this set the *Chebyshev fibre*. The semi-iterative method is the point with equal weights $w_j = 1/N$ (Section 2.3 and Corollary 7.1). All points of the fibre have the same residual polynomial $p_N^\star$ and differ in their intermediate ones.

By a classical argument [9, 27], such a method is the sequence of conjugate gradient coefficients recorded on a diagonal model quadratic whose eigenvalues are the roots of $p_N^\star$, started from an initial gradient with squared components $\propto w_j$, and replayed on the actual objective.

5.2 Factorization of the kernel

In Jacobi coordinates the kernel has the left–right structure of the Green function (3).

Theorem 5.2 (Jacobi first-variation kernel)**.** *Let $\mathcal{R}_J$ be the method of Theorem 5.1. In the variables $x = z(\lambda)$ (after the perturbation) and $y = z(\nu)$ (before the perturbation), $K_s(x,y) = -\frac{2}{\Delta P_N(\eta)} Q_{N-s-1}^{(s+1)}(x) P_s(y)$, $0 \leq s \leq N-1$, and therefore $A_N(J) = \frac{2}{\Delta P_N(\eta)} \sup_{x,y\in[-1,1]} \Big(\sum_{s=0}^{N-1} Q_{N-s-1}^{(s+1)}(x)^2 P_s(y)^2\Big)^{1/2}$.*

The leading polynomial $P_s(y)$ is the (unnormalized) residual of the ν-mode before step s; the trailing polynomial $Q_{N-s-1}^{(s+1)}(x)$ is the response of the remaining chain to an impulse in the λ-mode. For the semi-iterative method the leading polynomial is $P_0 = 1$, $P_s = 2^{1-s}T_s$ for $s \geq 1$, and the trailing one is $2^{s+1-N}U_{N-s-1}$; thus Theorem 5.2 reduces to Theorem 4.1.

5.3 The endpoint Green profile and a universal lower bound

On the Chebyshev fibre, evaluating the kernel at $x = y = 1$, i.e. at $\lambda = \nu = \mu$, and comparing with (3) at $x = 1$ yields the diagonal of the Green matrix $(I - J)^{-1}$:

$$K_s(1,1) = -\tfrac{2\epsilon_N^\star}{\Delta}\, d_s, \qquad d_s := \big[(I-J)^{-1}\big]_{ss} = \frac{P_s(1)\, Q_{N-s-1}^{(s+1)}(1)}{P_N(1)} > 0, \tag{5}$$

because $P_N(1)/P_N(\eta) = T_N(1)/T_N(\eta) = \epsilon_N^\star$. We call d the *endpoint Green profile*. For any final-exact realization set $d_s := -\Delta K_{s,\mathcal{R}}(\mu,\mu)/(2\epsilon_N^\star)$, allowing signs.

Lemma 5.1 (Green mass)**.** *For every final-Chebyshev-exact realization, $\sum_{s=0}^{N-1} d_s = N^2$.*

Proof. The identity $\sum_s K_s(\mu,\mu) = p_N^{\star\prime}(\mu) = -2\epsilon_N^\star N^2/\Delta$ from (T4) gives the claim. □

The gain is at least the endpoint value, and the endpoint value is the ℓ_2 norm of a profile with fixed sum. The Cauchy–Schwarz inequality therefore gives a universal lower bound. This observation is due to an anonymous referee.

Proposition 5.1 (Universal lower bound)**.** *For every final-Chebyshev-exact realization, $A_N(\mathcal{R}) \geq \frac{2\epsilon_N^\star}{\Delta}\Big(\sum_{s=0}^{N-1} d_s^2\Big)^{1/2} \geq \frac{2\epsilon_N^\star}{\Delta}\, N^{3/2}$.*

Proof. The first inequality evaluates the gain at (μ,μ). The second follows from $N^4 = (\sum_s d_s)^2 \leq N \sum_s d_s^2$ and Lemma 5.1. □

The problem of minimizing the gain over the Chebyshev fibre thus contains the problem of distributing a fixed Green mass N^2 along the chain as evenly as possible. Equality in the lower bound requires $d_s \equiv N$. For the semi-iterative method (5) and Theorem 4.1 give $d_0 = N$ and $d_s = 2(N-s)$ for $s \geq 1$: the mass is concentrated at early times, and $\sum_s d_s^2 \sim \frac{4}{3}N^3$, which is Theorem 4.1 again, with equality in the first inequality of Proposition 5.1.

Remark 5.1 (Momentum spreads the Green mass). For Chebyshev stepsizes the endpoint profile consists of the *eigenvalues* $(1-\zeta_j)^{-1}$ of $(I-J)^{-1}$ (proof of Theorem 3.1), on the Chebyshev fibre of its *diagonal*, which they majorize (Schur): $N^3 \leq \sum_s d_s^2 \leq \sum_j (1-\zeta_j)^{-2} \sim \frac{2}{3}N^4$.

6 The Sine–Jacobi Method

6.1 Choice of the weights

The fibre carries a symmetry: time reversal $J \mapsto RJR$, where R reverses the order of the basis, preserves the spectrum. Reversing the semi-iterative method gives the *reversed Chebyshev* method (for $N \geq 2$, $a_{N-1}^2 = \frac{1}{2}$, all other interior $a_s^2 = \frac{1}{4}$), whose profile $d_s = 2(s+1)$ for $s < N-1$, $d_{N-1} = N$, is the mirror image of d^{pref}. The normalized eigenvectors v_j of a Jacobi matrix satisfy $(v_j)_0 (v_j)_{N-1} = \prod_s a_s / P_N'(\zeta_j)$: by the cofactor formula the $(0, N-1)$ entry of $(xI-J)^{-1} = \sum_j v_j v_j^\top/(x-\zeta_j)$ is $\prod_s a_s / P_N(x)$, and the residues at ζ_j agree. By (T4), $|P_N'(\zeta_j)| = 2^{1-N} N / \sin\theta_j$, $\theta_j = (2j-1)\pi/(2N)$. Hence reversal maps the weights w to $w_j^{\mathrm{rev}} \propto \sin^2\theta_j / w_j$: the equal weights go to $\sin^2\theta_j$, and the only fixed point is $w_j^{\sin} = \sin\theta_j / \sum_k \sin\theta_k$. We call the corresponding persymmetric Jacobi matrix $J_N^{\sin}$ the *sine–Jacobi matrix*.

Lemma 6.1 (Time reversal). *$K_s^{RJR}(x,y) = K_{N-1-s}^J(y,x)$ and $d_s^{RJR} = d_{N-1-s}^J$. Hence A_N, $\sum_s d_s^2$, and Γ of Corollary 8.1(b) with isotropic e_0 are invariant under reversal, and every differentiable reversal-invariant function on the fibre has a critical point at $J_N^{\sin}$.*

Proof. The leading polynomials of RJR are the trailing ones of J and conversely, so Theorem 5.2 gives the first identity and (5) the second; the coupling of (λ, ν) in Γ of Corollary 8.1(b) is symmetric. In the coordinates $v_j = \log w_j$, modulo constants, reversal σ is the point reflection $v \mapsto 2v^{\sin} - v$, so $F \circ \sigma = F$ forces $\nabla F(v^{\sin}) = -\nabla F(v^{\sin})$. □

On the path $w_j \propto \sin^{2t}\theta_j$, which reversal maps to $t \mapsto 1-t$, the function $\sum_s d_s^2/N^3$ at $N = 64$ equals 1.318, 1.174, 1.142 for $t = 0, \frac{1}{4}, \frac{1}{2}$; whether the critical point is the global minimum on the fibre is open, see the lower bound in [24].

Theorem 6.1 (The sine–Jacobi matrix). *$J_N^{\sin}$ is persymmetric, has zero diagonal,* $\det(xI - J_N^{\sin}) = 2^{1-N} T_N(x)$, *and squared off-diagonal entries* $a_{s,N}^2 = \frac{1}{4} \frac{\sin^2(\pi s/N)}{\sin\big(\pi(s-\frac{1}{2})/N\big)\sin\big(\pi(s+\frac{1}{2})/N\big)}$, $\quad s = 1, \ldots, N-1$; *uniformly for $t = s/N$ in compact subsets of $(0,1)$, $a_{s,N}^2 = \frac{1}{4} + \frac{\pi^2}{16N^2\sin^2(\pi t)} + O(N^{-4})$. Its leading polynomials satisfy $|P_s(x)| \leq P_s(1)$ on $[-1,1]$, $s \leq N-1$, and so do all its principal blocks.*

The squared coefficients are symmetric, $a_s^2 = a_{N-s}^2$, and close to the Chebyshev value $\frac{1}{4}$ in the interior. The leading polynomials are a root-of-unity specialization of the Rogers polynomials [14, 29]: explicit convolutions with nonnegative coefficients in $\cos k\theta$, which gives the maximum principle. The proof (Appendix D) verifies this directly. With $p_0 = 1$, $p_1 = \eta$, $p_{s+1} = \eta p_s - a_{s,N}^2 p_{s-1}$ (so that $p_s = P_s(\eta) > 0$), Theorem 5.1 gives the *sine–Jacobi method*: $x_1 = x_0 - \frac{2}{L+\mu}\nabla f(x_0)$ and

$$x_{s+1} = \tfrac{\eta p_s}{p_{s+1}} x_s - \tfrac{a_{s,N}^2 p_{s-1}}{p_{s+1}} x_{s-1} - \tfrac{2p_s}{\Delta p_{s+1}} \nabla f(x_s), \qquad 1 \leq s \leq N-1. \tag{6}$$

It has the same cost per step as the semi-iterative method and returns the same x_N on every quadratic; since the first two coefficients differ by 1, it is a heavy-ball-type method with momentum $a_{s,N}^2 p_{s-1}/p_{s+1}$ and stepsize $2p_s/(\Delta p_{s+1})$.

6.2 The gain of the sine–Jacobi method

The asymptotics of the endpoint Green profile is described by a continuum profile. Let $f(t) := \frac{2}{\pi^{3/2}}\sqrt{\sin \pi t}\,\mathrm{K}\Big(\sin\frac{\pi t}{2}\Big)$, $0 < t < 1$, where $\mathrm{K}(k) = \int_0^{\pi/2}(1-k^2\sin^2\theta)^{-1/2}\,d\theta$ is the complete elliptic integral of the first kind [22, Ch. 19]. It is the solution of $f'' + \frac{\pi^2}{4\sin^2(\pi t)} f = 0$ with $f(t) \sim \sqrt{t}$ as $t \downarrow 0$ (Theorem D.1). Set $\varphi_{\sin}(t) := f(t)f(1-t)/\int_0^1 f(u)f(1-u)\,du$ and $c_{\sin} := \int_0^1 \varphi_{\sin}^2$. Numerically $1.14269215 < c_{\sin} < 1.14269221$; this enclosure is formally verified [23].

Theorem 6.2 (Sine–Jacobi profile and gain)**.** *Let d_s be the endpoint Green profile of $J_N^{\sin}$ and $\varphi_N(t) := d_{\lfloor tN\rfloor}/N$, $0 \le t < 1$. Then $\varphi_N \to \varphi_{\sin}$ in $L^2(0,1)$, the sine–Jacobi method is final-Chebyshev-exact, and $A_N(J_N^{\sin}) = \frac{2\epsilon_N^\star}{\Delta}(\sum_s d_s^2)^{1/2} \sim 2\sqrt{c_{\sin}}\,\frac{N^{3/2}}{\Delta}\,\epsilon_N^\star$, where $2\sqrt{c_{\sin}} \approx 2.137936$.*

By the maximum principle of Theorem 6.1 and persymmetry, the supremum in Theorem 5.2 is attained at $x = y = 1$, so the gain is the ℓ_2 norm of the profile (Appendix D). The constant improves on $4/\sqrt{3} \approx 2.3094$ by 7.4% (14.3% on the squared scale of Remark 8.1) and exceeds the lower bound 2, which would require $d_s \equiv N$, by 7%.

7 Beyond First Order: Finite Drift and Nonquadratic Objectives

The gain A_N describes the linear response. We now control the full response to drift of finite size, which gives convergence guarantees for nonquadratic objectives. For a Jacobi matrix J and $0 \le a \le b+1 \le N$, let $\widetilde{D}_{a:b}(x) := 2^{\,b-a+1}\det(xI - J_{a:b})$, $\widetilde{D}_{a:a-1} := 1$, and $g_{a:b} := \max_{[-1,1]}|\widetilde{D}_{a:b}|$. The factor 2^{b-a+1} is a normalization: for zero diagonal and all couplings $\frac{1}{2}$, $\widetilde{D}_{a:b} = U_{b-a+1}$.

Theorem 7.1 (Finite Hessian drift in Jacobi coordinates)**.** *Let $\mathcal{R}_J$ be a final-Chebyshev-exact Jacobi method, $\mu I \preceq H \preceq LI$, $Z := z(H)$, and let $E = (E_0, \dots, E_{N-1})$ with $\delta := \max_s \|E_s\|$.*

(i) The output operator has the exact expansion $\mathcal{T}_{\mathcal{R}_J}(H+E) = \sum_{k=0}^N \mathcal{T}^{(k)}$, where

$$\mathcal{T}^{(k)} = \frac{\epsilon_N^\star}{2}\Big(-\frac{4}{\Delta}\Big)^k \sum_{s_1<\cdots<s_k} \widetilde{D}_{s_k+1:N-1}(Z)\,E_{s_k}\,\widetilde{D}_{s_{k-1}+1:s_k-1}(Z) \cdots E_{s_1}\,\widetilde{D}_{0:s_1-1}(Z),$$

$\mathcal{T}^{(0)} = p_N^\star(H)$ and $\mathcal{T}^{(1)} = \mathcal{T}'_{\mathcal{R}_J,H}[E]$.

(ii) For every k, $\|\mathcal{T}^{(k)}\| \le \frac{\epsilon_N^\star}{2}\big(\frac{4\delta}{\Delta}\big)^k G_k(J)$, where $G_k(J)$ is the sum over $s_1 < \cdots < s_k$ of $g_{0:s_1-1}g_{s_1+1:s_2-1}\cdots g_{s_k+1:N-1}$.

(iii) Always $G_k(J) \ge 2^{1-k}[t^k]\,T_N(1+t)$, with equality for every k if each $\widetilde{D}_{a:b}$ attains its maximum on $[-1,1]$ at $x = 1$. In that case, with $\mathcal{T} = \mathcal{T}_{\mathcal{R}_J}(H+E)$, $\mathcal{T}' = \mathcal{T}'_{\mathcal{R}_J,H}[E]$ and $u = 2\delta/\Delta$,

$$\|\mathcal{T} - p_N^\star(H)\| \le \epsilon_N^\star(T_N(1+u)-1), \qquad \|\mathcal{T}'\| \le \epsilon_N^\star N^2 u,$$
$$\|\mathcal{T} - p_N^\star(H) - \mathcal{T}'\| \le \epsilon_N^\star(T_N(1+u) - 1 - N^2u) \le \tfrac{1}{6}\,\epsilon_N^\star N^4u^2\cosh(N\sqrt{2u}).$$

(iv) Conversely, for every final-Chebyshev-exact realization $\mathcal{R}$ (of any memory), if $\mu \in \mathrm{spec}(H)$ and $E_s = -\delta I$ for all s, then the three Chebyshev bounds in (iii) hold in reverse.

The terminal factor $\epsilon_N^\star T_N(1+u)$ equals $\max_{[\mu-\delta,L+\delta]}|p_N^\star|$: under the block-maximum condition in (iii), time-varying drift of size δ is, at every order, no worse than the static shift $H \mapsto H - \delta I$, and by (iv) no Chebyshev-exact method does better.

Corollary 7.1 (Finite drift for the two momentum methods)**.** *(a) The semi-iterative method is the Jacobi method with $b_s = 0$, $a_1^2 = \frac{1}{2}$, $a_s^2 = \frac{1}{4}$ ($s \ge 2$); its blocks are $\widetilde{D}_{0:b} = 2T_{b+1}$ and $\widetilde{D}_{a:b} = U_{b-a+1}$ ($a \ge 1$). (b) Every block of $J_N^{\sin}$ has nonnegative Fourier coefficients in θ, $x = \cos\theta$. Both methods therefore satisfy Theorem 7.1(iii); its three Chebyshev bounds are sharp.*

The two momentum methods thus have the same, optimal, worst-case response; they differ in the ℓ_2 accumulation measured by A_N (Section 8).

Objectives with Lipschitz Hessian. Let f be twice differentiable near a local minimizer $x^\star$, $H_\star := \nabla^2 f(x^\star)$ satisfies $\mu I \preceq H_\star \preceq LI$, and $e_s := x_s - x^\star$. For a method that queries the gradient at its iterates, $\nabla f(x_s) = \bar{H}_s e_s$ with the averaged Hessian $\bar{H}_s := \int_0^1 \nabla^2 f(x^\star + te_s)\, dt$. The error after N steps is therefore exactly $e_N = \mathcal{T}_{\mathcal{R}}(\bar{H}_0, \dots, \bar{H}_{N-1})\, e_0$: a smooth objective is a quadratic with drifting Hessian, and the drift $\bar{H}_s - H_\star$ is controlled by the Lipschitz constant ρ of $\nabla^2 f$. If $\nabla^2 f$ is ρ-Lipschitz on a convex set containing $x^\star$ and iterates with $\|e_s\| \le R$, $s < N$, then $E_s := \bar{H}_s - H_\star$ satisfies $\|E_s\| \le \rho R/2$, and Corollary 7.1 gives $\|e_N\| \le \epsilon_N^\star T_N(1+\rho R/\Delta)\|e_0\|$ for both methods. The a priori bound R can be removed: both methods keep their iterates in the ball of radius $\|e_0\|$ as long as the drift stays below μ. This gives a local complexity bound.

Theorem 7.2 (Local linear convergence)**.** *Let $\nabla^2 f$ be ρ-Lipschitz on the ball $\mathbb{B} := \{x : \|x - x^\star\| \le r\}$, where $\rho r \le 2\mu$. Run the semi-iterative or the sine–Jacobi method, calibrated on $[\mu, L]$, from $x_0 \in \mathbb{B}$. Then $\|e_s\| \le \|e_0\|$ for $0 \le s \le N$, and $\|e_N\| \le \epsilon_N^\star\, T_N\Big(1 + \frac{\rho r}{\Delta}\Big)\|e_0\| \le 2\exp\Big(-\frac{2N}{\sqrt{\kappa}}\Big(1 - \sqrt{\frac{\rho r}{2\mu}}\Big)\Big)\|e_0\|$. In particular, if $\rho\|e_0\| \le \mu/2$, restarting the method with horizon $N = \lceil \sqrt{\kappa} \ln 4 \rceil$ at its output halves the error per block. Any target $0 < \varepsilon < \|e_0\|$ is reached within $N\lceil \log_2(\|e_0\|/\varepsilon)\rceil$ gradient calls.*

The proof (Appendix E) applies the expansion of Theorem 7.1 to the leading blocks $J_{0:t-1}$, by induction over t. Its second inequality shows that drift of size $\delta = \rho r/2 < \mu$ acts as the condition number $\kappa/(1 - \sqrt{\delta/\mu})^2$: drift negligible relative to L but a fraction of μ changes the final error substantially, and by Theorem 7.1(iv) no Chebyshev-exact method does better.

Theorem 7.2 is uniform in time: it bounds every drift by its largest value $\rho r/2$, and in that form the two methods are indistinguishable. Along a trajectory, however, the drift decays with the error, as $\|\bar{H}_s - H_\star\| \le \rho\|e_s\|/2$, and then the realization matters already at the first nonlinear order. The relevant quantity is the endpoint Green profile weighted by the residual decay, $C_N(J) := \sum_{s=0}^{N-1} d_s\, m_s$, $\quad m_s := \frac{P_s(1)}{P_s(\eta)} = \max_{[\mu,L]} |r_s|$, where the last equality uses $|P_s(x)| \le P_s(1)$ on $[-1, 1]$. Since $m_s \le 1$, $C_N(J) \le \sum_s d_s = N^2$.

Theorem 7.3 (Second-order error along the trajectory)**.** *Under the assumptions of Theorem 7.2, let $r_0 := \|e_0\|$. For the semi-iterative and the sine–Jacobi method, $\|e_N - p_N^\star(H_\star)e_0\| \le \frac{\rho}{\Delta}\,\epsilon_N^\star\, C_N(J)\, e^{2N^2\rho r_0/\Delta}\, r_0^2$. The constant $C_N(J)$ is sharp: for $X = \mathbb{R}$, $f(x) = \frac{\mu}{2}x^2 - \frac{\rho}{6}x^3$ and $x_0 = r_0 > 0$, $e_N - p_N^\star(\mu)e_0 = \frac{\rho}{\Delta}\,\epsilon_N^\star\, C_N(J)\, r_0^2\,(1 + O(r_0))$, $\quad r_0 \downarrow 0$.*

The exponential factor is at most e^2 while $\rho r_0 \le \Delta/N^2$, the scale on which the first nonlinear term dominates; the uniform bound of Theorem 7.2 corresponds to $C_N(J)$ replaced by N^2.

Remark 7.1 (Size of C_N). For the semi-iterative method $d_0 = N$, $d_s = 2(N - s)$ for $1 \le s < N$, and $m_s = 1/T_s(\eta) = 1/\cosh(sa)$, $a \sim 2/\sqrt{\kappa}$, so a Riemann sum gives $C_N^{\mathrm{pref}}/(N\sqrt{\kappa}) \to \int_0^{2\tau}(1 - \frac{x}{2\tau})\operatorname{sech} x\, dx$ as $\kappa \to \infty$ with $N/\sqrt{\kappa} \to \tau > 0$; this increases to $\pi/2$ as $\tau \to \infty$: the scale N^2 becomes $N\sqrt{\kappa}$. The limiting normalized sine–Jacobi profile vanishes at the initial time, where the drift is largest, and $C_N^{\sin}/C_N^{\mathrm{pref}}$ depends, to three digits, only on τ: 0.926, 0.611, 0.329 for $\tau = \ln 4$ (restarts of Theorem 7.2), 4, 10.

8 Two Models of Hessian Variation

Sections 3–7 measure the worst case. In applications the variation is usually proportional to the current error and spread over many pairs of eigenvalues. For two standard models, a smooth nonquadratic objective and gossip averaging with random link failures, the leading error term is an explicit spectral expression, computable before the method is run (Section 9 tests it).

8.1 Perturbed runs

All methods of this paper have the form $x_{s+1} = \alpha_s x_s - \beta_s x_{s-1} - \gamma_s \nabla f(x_s)$ with $\alpha_s - \beta_s = 1$ and $\beta_0 = 0$: the Jacobi methods of Theorem 5.1 and gradient descent with Chebyshev stepsizes ($\beta_s = 0$). For $0 \le s < t \le N$ let $G_{t,s+1}$ be the tail function of Lemma A.1 with terminal time t, and write

$G_s := G_{N,s+1}(H)$. By Lemma A.1 the first-order kernel is $K_s(\lambda,\nu) = -\gamma_s\, G_s(\lambda)\, r_s(\nu)$. A *perturbed run* replaces the gradient by an oracle output $\hat{g}_s = He_s + \eta_s$ and may perturb the update itself: $x_{s+1} = \alpha_s x_s - \beta_s x_{s-1} - \gamma_s \hat{g}_s + \xi_s$. Hessian drift is the case $\eta_s = E_s e_s$, $\xi_s = 0$.

Lemma 8.1 (Perturbed runs)**.** *For arbitrary η_s, ξ_s, possibly random and depending on the trajectory, the errors of the perturbed run satisfy $e_t = r_t(H)e_0 + \sum_{s<t} G_{t,s+1}(H)\,(\xi_s - \gamma_s\eta_s), \quad 0 \le t \le N$.*

Proof. The difference $e_t - r_t(H)e_0$ solves the unperturbed recurrence with zero initial data and the forcing $\xi_s - \gamma_s\eta_s$ added at time $s+1$. By linearity it is the sum of the responses to the individual forcings, and the response at time t to a forcing v at time $s+1$ is $G_{t,s+1}(H)v$ by the definition of the tail functions. □

For random perturbations the summands are martingale differences, which gives an exact identity for the mean squared error.

Proposition 8.1 (Random perturbations)**.** *Let $\mathcal{F}_s$ be generated by η_r, ξ_r, $r < s$. Assume that $\mathbb{E}[\eta_s \mid \mathcal{F}_s] = 0$, $\mathbb{E}[\xi_s \mid \mathcal{F}_s, \eta_s] = 0$, and that all second moments are finite. Then $\mathbb{E}e_t = r_t(H)e_0$ for every t, and $\mathbb{E}\|e_N - p_N(H)e_0\|^2 = \sum_s(\gamma_s^2\mathbb{E}\|G_s\eta_s\|^2 + \mathbb{E}\|G_s\xi_s\|^2)$. If, moreover, $\xi_s = 0$ and the conditional covariance of η_s given $\mathcal{F}_s$ is $\sigma^2 C_s(e_s)$, where $C_s(v)$ is positive semidefinite and homogeneous quadratic in v, then, with $q_s(v) := \operatorname{tr}\big(G_s C_s(v) G_s\big)$ and $e_s^0 := r_s(H)e_0$, $\mathbb{E}\|e_N - p_N(H)e_0\|^2 = \sigma^2\,\Gamma + R, \quad \Gamma := \sum_{s=0}^{N-1}\gamma_s^2\, q_s(e_s^0)$, where $0 \le R = O(\sigma^4)$ as $\sigma \to 0$.*

Proof. Taking expectations in Lemma 8.1 gives $\mathbb{E}e_t = r_t(H)e_0$. For $r < s$, the summand with index r is $\mathcal{F}_s$-measurable and the one with index s has zero conditional mean, so the cross terms of $\mathbb{E}\|\sum_s G_s(\xi_s - \gamma_s\eta_s)\|^2$ vanish; within one step, $\mathbb{E}\langle G_s\eta_s, G_s\xi_s\rangle = 0$ because ξ_s has zero mean given η_s. For the second claim, $\mathbb{E}[\|G_s\eta_s\|^2 \mid \mathcal{F}_s] = \sigma^2 q_s(e_s)$, and the quadratic form $q_s \ge 0$ satisfies $\mathbb{E}q_s(e_s) = q_s(\mathbb{E}e_s) + \mathbb{E}q_s(e_s - \mathbb{E}e_s)$. The first term is $q_s(e_s^0)$. The second is nonnegative and at most a constant times $\mathbb{E}\|e_s - e_s^0\|^2$, which by the first claim at time s equals $\sigma^2\sum_{r<s}\gamma_r^2\,\mathbb{E}\operatorname{tr}\big(G_{s,r+1}C_r(e_r)G_{s,r+1}\big)$ and is $O(\sigma^2)$ by induction on s. □

Γ needs one eigendecomposition of H and $O(N)$ polynomial values; for e_0 uniform on the sphere its mean is the average over eigenvectors.

Remark 8.1 (Stochastic curvature). Let the gradient at step s be $(H + E_s)e_s$, with independent random E_s, $\mathbb{E}E_s = 0$, $\mathbb{E}\|E_s\|_{\mathrm{HS}}^2 < \infty$, and let e_0 be independent of them and uniform on the unit sphere of $X = \mathbb{R}^d$. Then $\eta_s = E_s e_s$, $C_s(v) = \mathbb{E}E_s vv^\top E_s^\top$, and since $\gamma_s G_s(H)E_s r_s(H) = -\sum_{\lambda,\nu} K_s(\lambda,\nu)\Pi_\lambda E_s \Pi_\nu$ (Lemma A.1) and $\mathbb{E}\|Me_0\|^2 = \|M\|_{\mathrm{HS}}^2/d$, the leading term is $\frac{1}{d}\sum_{s,\lambda,\nu} K_s(\lambda,\nu)^2\,\mathbb{E}\|\Pi_\lambda E_s \Pi_\nu\|_{\mathrm{HS}}^2$. If the E_s are identically distributed, it is at most $\frac{A_N^2}{d}\mathbb{E}\|E_0\|_{\mathrm{HS}}^2$, with equality when E_0 couples only a pair at which the supremum defining A_N is attained, (μ,μ) for both momentum methods. So A_N^2 is the worst-case coefficient of the curvature-noise variance, and asymptotically the sine–Jacobi method has a 14% smaller worst-case leading excess than the semi-iterative method (Theorems 4.1 and 6.2).

8.2 Where the gain comes from

In both models below perturbations scale with the current error or its square, so the leading terms weight their transmission by the unperturbed residuals. The following result isolates the first step.

Proposition 8.2 (First-step transmission)**.** *For every final-Chebyshev-exact Jacobi method, the first-gradient kernel at the slowest mode is input-independent: $K_0(\mu,\nu) = -\frac{2\epsilon_N^\star}{\Delta}\,d_0$ for all $\nu \in [\mu, L]$. For the semi-iterative method $d_0 = N$. For the sine–Jacobi method $d_0 = \sin\frac{\pi}{2N}\sum_{j=1}^{N}\cot\frac{\theta_j}{2} = \ln N + \gamma_{\mathrm{E}} + \ln\frac{8}{\pi} + O(\frac{\ln N}{N^2})$, with Euler's constant γ_{E}, and $d_{N-1} = d_0$ by persymmetry.*

The proof is in Appendix F. At output mode μ, the first-step coefficients d_0 are N and $\ln N + O(1)$, independently of the input mode. For long blocks this relative reduction far exceeds the asymptotic gain improvement of about 7%. Errors independent of the current error, such as inexact-oracle errors [7], enter Lemma 8.1 without the factor $r_s(H)e_0$ and therefore have different trajectory weights.

8.3 Two models

Corollary 8.1 (Leading terms)**.** *Let* $e_s^0 = r_s(H)e_0$.

(a) Smooth energies. *If* f *is three times differentiable at a stationary point* $x^\star$*, with* $H = \nabla^2 f(x^\star)$ *and* $\mu I \preceq H \preceq LI$*, then* $e_N - p_N(H)e_0 = Z(e_0) + o(\|e_0\|^2)$ *with*

$$Z(v) = -\tfrac{1}{2}\textstyle\sum_s \gamma_s\, G_s\, \nabla^3 f(x^\star)\big[r_s(H)v,\, r_s(H)v\big],$$

and, if $\nabla^2 f$ *is* ρ*-Lipschitz near* $x^\star$*,* $\|Z(v)\| \le \frac{\rho}{\Delta}\epsilon_N^\star C_N(J)\|v\|^2$ *for the two momentum methods (Theorem 7.3).*

(b) Gossip with link failures. *Let* $\mathcal{L} = \sum_e w_e b_e b_e^\top$*,* $b_e = \mathbf{1}_i - \mathbf{1}_j$*, be the Laplacian of a connected graph, and let* $[\mu, L]$*,* $0 < \mu < L$*, enclose its nonzero spectrum; the methods applied to* $f(x) = \frac{1}{2}x^\top \mathcal{L} x$ *are accelerated gossip iterations [20, 28]. Random link failures in consensus are studied in [13]. Here each edge is active independently across edges and steps with probability* $1-p$*,* $0 \le p < 1$*, and we divide active weights by* $1-p$ *so that the expected Laplacian is* $\mathcal{L}$*. The average is preserved; with* $x^\star = \bar{x}_0\mathbf{1}$*,* $e_s \perp \mathbf{1}$*, and* $\mathbb{E}\|e_N - p_N(\mathcal{L})e_0\|^2 = \frac{p}{1-p}\,\Gamma + O(p^2)$ *with* $\Gamma = \sum_s \gamma_s^2 \sum_e w_e^2\,(b_e^\top e_s^0)^2\,\|G_s b_e\|^2$.

The proofs are in Appendix F. Since $\mathbb{E}e_N = p_N(\mathcal{L})e_0$ (Proposition 8.1), in (b) the mean squared error is $\mathbb{E}\|e_N\|^2 = \|p_N(\mathcal{L})e_0\|^2 + \frac{p}{1-p}\Gamma + O(p^2)$. Model (b) has the form $\eta_s = E_s e_s$ of Remark 8.1 on $\mathbf{1}^\perp$, with $E_s = \sum_e (\chi_{s,e}/(1-p) - 1) w_e b_e b_e^\top$ for the link indicators $\chi_{s,e}$; the same coefficient governs unrescaled weights with the interval $(1-p)[\mu, L]$.

9 Numerical Experiments

Protocol. We compare four realizations of $p_N^\star$ with one gradient per step: the semi-iterative method, the sine–Jacobi method, the Jacobi method with $w_j \propto \sin^{1/2}\theta_j$ (the point $t = \frac{1}{4}$ of the path of Lemma 6.1, a further member of the Chebyshev fibre), and gradient descent with Chebyshev stepsizes in the fractal order of Lebedev and Finogenov [15, 1] ($N = 2^k$). Without Hessian variation all four return the same x_N. We therefore report the error after one block of N steps relative to the error of the same block without variation (Figure 1): the value 1 means that the variation does not matter. The predictions are those of Corollary 8.1, computed from the spectrum of the reference Hessian before any simulation. Parameters and code are available at `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/chebyshev1-code`.

Smooth energy. Let $f(u) = \frac{1}{2}u^\top A u + \frac{1}{4}\sum_i u_i^4 - 100\sum_i u_i$, where A is the Dirichlet Laplacian on 63 interior points of $(0,1)$, the discretization of $-u'' + u^3 = 100$. Then $\|u^\star\| = 27.9$, $\kappa(H_\star) = 283$, $\nabla^3 f(u^\star)[v,v] = 6\,u^\star \circ v \circ v$, and a block of $N = 64$ steps reduces the error of the quadratic model by a factor of about 10^3. We use 21 random directions v. For $r_0 \le 1$ the three Jacobi methods stay within 3% of the quadratic model (Figure 1(a)); gradient descent exceeds it by at least 20% at $r_0 \approx 0.02$ and diverges for $r_0 \ge 1.2$ (in some directions from $r_0 = 0.4$). For $r_0 \ge 10$ the errors are ordered as predicted in every direction: sine–Jacobi, $w_j \propto \sin^{1/2}\theta_j$, semi-iterative; at $r_0 = 30$, slightly more than $\|u^\star\|$, they exceed the error of the quadratic model by factors 1.02–2.1, 1.3–3.3 and 4.1–7.3. At intermediate r_0 the second-order term can partly cancel the linear one, and the ordering depends on the direction. Up to $r_0 \approx 10$ the second-order prediction is accurate within 6% for the Jacobi methods. As N grows, the second-order term of the sine–Jacobi method becomes smaller relative to the semi-iterative one: the ratio of $\|Z(v)\|$ decreases from 0.40 at $N = 16$ to 0.063 at $N = 256$ for a random direction, and for the slowest eigenvector it equals $C_N^{\sin}/C_N^{\mathrm{pref}}$ of Remark 7.1 to three digits.

Gossip with link failures. We use four graphs with unit weights: a ring of six 8-cliques (48 nodes, $\kappa = 99$), a barbell of two 12-cliques (24 nodes, $\kappa = 96$), a random geometric graph (60 nodes, radius 0.25, $\kappa = 60$) and the 8×8 grid ($\kappa = 51$), $N = 32$, and e_0 uniform on the unit

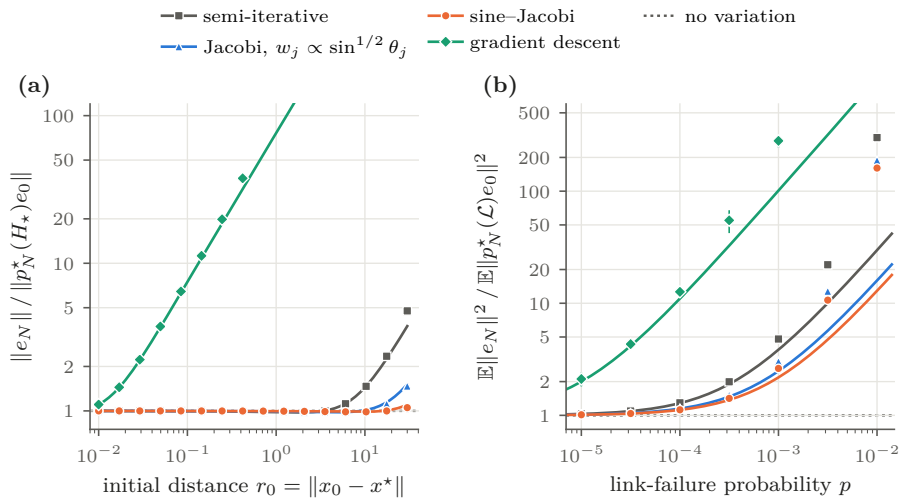


Figure 1: Error after one block relative to the same block without Hessian variation (dotted line), which is the same for all four methods. Lines: predictions of Corollary 8.1; markers: measurements. (a) Smooth energy, $N = 64$, $x_0 = u^\star + r_0 v$ for a random unit v; $\|e_N\|/\|p_N^\star(H_\star)e_0\|$ against r_0, with the second-order prediction $\|p_N^\star(H_\star)v + r_0 Z(v)\|/\|p_N^\star(H_\star)v\|$. Values slightly below 1 occur because $Z(v)$ is partly opposite to $p_N^\star(H_\star)v$. (b) Gossip on the ring of cliques, $N = 32$: mean squared error against the link-failure probability, 10^5 runs per marker (95% intervals), with the first-order prediction $1 + \frac{p}{1-p}\Gamma/\mathbb{E}\|p_N^\star(\mathcal{L})e_0\|^2$.

sphere of $\mathbf{1}^\perp$ in each run; the methods share the random numbers. The first-order prediction is accurate within 8% for the Jacobi methods for $p \le 3 \cdot 10^{-4}$, and within 15% for gradient descent for $p \le 10^{-4}$, on all four graphs; for larger p it underestimates the error, by 4–36% at $p = 10^{-3}$ for the Jacobi methods (Figure 1(b)). On every graph and at every $p \in [10^{-5}, 10^{-1}]$ the methods are ordered as predicted: sine–Jacobi, $w_j \propto \sin^{1/2}\theta_j$, semi-iterative, and, far behind, gradient descent. At $p = 10^{-3}$ the failures increase the mean squared error by factors 4.8, 4.6, 1.9, 1.9 on the four graphs for the semi-iterative method, 2.6, 3.1, 1.6, 1.2 for the sine–Jacobi method, and 43–281 for gradient descent. Extrapolated to $p \to 0$, the measured coefficients for $N = 8, 16, 24, 32$ agree with Γ within 11% (median 1.8%), and $\Gamma^{\sin}/\Gamma^{\mathrm{pref}}$ decreases from 0.71–0.82 at $N = 8$ to 0.26–0.63 at $N = 32$.

10 Conclusions

Realizations of the same Chebyshev polynomial respond differently to a changing Hessian, as measured by an explicit gain. Momentum is necessary for a small gain, and the sine–Jacobi method, the reversal-symmetric member of the positive two-term class, comes within seven percent of a universal lower bound. Beyond first order both momentum methods are optimal in the worst case and differ, predictably, where the variation is not uniform in time. Open questions are the global optimum over two-term methods, longer memory, and other final polynomials.

Declarations. The authors have no competing interests to declare that are relevant to the content of this article. The accompanying Lean 4 development [23] formalizes the realization expansion and gain bounds, the prefix-exact recurrence and its uniqueness, the Jacobi kernel and finite-drift identities, and the sine–Jacobi spectral construction, endpoint-profile limit, gain asymptotics, and certified enclosure of $c_{\sin}$. Experimental details and code are given in Section 9.

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A Proofs for Section 3

Lemma 3.1. Let $\mathsf{C}_s(A) := M_s \otimes I_X - g_s q_s^\top \otimes A$, so that $\mathsf{C}_s(H + \delta B_s) = \mathsf{C}_s(H) - \delta\, g_s q_s^\top \otimes B_s$. Every factor of (4) is affine in δ, and the product rule gives $\mathcal{T}'_{\mathcal{R},H}[B] = \sum_s \mathcal{T}'_s(B_s)$ with

$$\mathcal{T}'_s(B_s) = -(\ell^\top \otimes I_X)\, \mathsf{C}_{N-1}(H) \cdots \mathsf{C}_{s+1}(H)\, (g_s q_s^\top \otimes B_s)\, \mathsf{C}_{s-1}(H) \cdots \mathsf{C}_0(H)\, (b \otimes I_X).$$

If $Hx = \rho x$, then $\mathsf{C}_j(H)(y \otimes x) = (C_j(\rho)y) \otimes x$. Hence, for x in the ν-eigenspace, the factors to the right of position s give $(C_{s-1}(\nu) \cdots C_0(\nu) b) \otimes x$, the insertion gives $(g_s q_s^\top C_{s-1}(\nu) \cdots C_0(\nu) b) \otimes B_s x$, and after projection onto the λ-eigenspace the factors to the left act with parameter λ. Applying $\ell^\top \otimes I_X$ gives $\Pi_\lambda \mathcal{T}'_s(B_s)x = K_{s,\mathcal{R}}(\lambda, \nu)\Pi_\lambda B_s x$. □

Lemma A.1 (Prefix–insertion–tail factorization)**.** *Consider a two-term method $x_{t+1} = \alpha_t x_t - \beta_t x_{t-1} - \gamma_t \nabla f(x_t)$, $0 \le t \le N-1$, with $\beta_0 = 0$ and residual polynomials r_t. For $0 \le s \le N-1$ and real λ, let $G_{N,s+1}(\lambda)$ be the value u_N of the scalar recurrence $u_{t+1} = (\alpha_t - \gamma_t\lambda)u_t - \beta_t u_{t-1}$, $t = s+1, \dots, N-1$, started from $u_s = 0$, $u_{s+1} = 1$. Then $K_s(\lambda, \nu) = -\gamma_s\, G_{N,s+1}(\lambda)\, r_s(\nu)$.*

Proof. With $H_t = H + \delta B_t$ and $e_t = x_t - x^\star$, $e_{t+1} = (\alpha_t - \gamma_t H_t)e_t - \beta_t e_{t-1}$. The derivative e'_t at $\delta = 0$ satisfies $e'_0 = 0$ and $e'_{t+1} = (\alpha_t - \gamma_t H)e'_t - \beta_t e'_{t-1} - \gamma_t B_t e_t$, with $e_t = r_t(H)e_0$. By linearity e'_N is the sum of the responses to the forcings $-\gamma_s B_s e_s$ entered at time $s+1$; the homogeneous recurrence acts on the λ-eigenspace as the scalar one, so the response at time N to the data $(0, v)$ at times $(s, s+1)$ is $G_{N,s+1}(\lambda)v$. Thus $\Pi_\lambda e'_N = -\sum_s \gamma_s G_{N,s+1}(\lambda)\Pi_\lambda B_s r_s(H)e_0$, which for e_0 in the ν-eigenspace is $-\sum_s \gamma_s G_{N,s+1}(\lambda) r_s(\nu)\Pi_\lambda B_s e_0$; comparison with Lemma 3.1 gives the formula. □

Proposition 3.1. Since $\mathsf{C}_s(H + E_s) = \mathsf{C}_s(H) - g_s q_s^\top \otimes E_s$, expanding the ordered product gives a sum over subsets $S \subseteq \{0, \dots, N-1\}$ of $|S|$-linear terms in which the factors with $s \in S$ are replaced by $-g_s q_s^\top \otimes E_s$. The term $S = \varnothing$ is $p_{\mathcal{R},N}(H)$, the terms $|S| = 1$ form $\mathcal{T}'_{\mathcal{R},H}[E]$, and the others form $\mathcal{T}^{(\geq 2)}_{\mathcal{R},H}[E]$. By Lemma 3.1 and the Cauchy–Schwarz inequality, $\|\Pi_\lambda \mathcal{T}'_{\mathcal{R},H}[E]\Pi_\nu\|^2_{\mathrm{HS}} \leq \sum_s K_{s,\mathcal{R}}(\lambda,\nu)^2 \sum_s \|\Pi_\lambda E_s \Pi_\nu\|^2_{\mathrm{HS}} \leq A_N(\mathcal{R})^2 \sum_s \|\Pi_\lambda E_s \Pi_\nu\|^2_{\mathrm{HS}}$, and summing over (λ, ν) with $\sum_{\lambda,\nu} \|\Pi_\lambda E \Pi_\nu\|^2_{\mathrm{HS}} = \|E\|^2_{\mathrm{HS}}$ gives the bound. The kernel is continuous, so the supremum is attained at some (λ_*, ν_*). For $H = \mathrm{diag}(\lambda_*, \nu_*)$ and $E_s = K_{s,\mathcal{R}}(\lambda_*, \nu_*)e_1 e_2^\top$ the $(1,2)$ entry of $\mathcal{T}'_{\mathcal{R},H}[E]$ is $\sum_s K_{s,\mathcal{R}}(\lambda_*, \nu_*)^2 = A_N^2$, while $(\sum_s \|E_s\|^2_{\mathrm{HS}})^{1/2} = A_N$. □

Theorem 3.1. For gradient descent $C_s(\lambda) = 1 - \gamma_s \lambda$, so $K_s(\lambda, \nu) = -\gamma_s \prod_{i>s}(1 - \gamma_i \lambda) \prod_{i<s}(1 - \gamma_i \nu)$. Exactness means $\{\gamma_s^{-1}\} = \{\lambda_j\}$; let $\gamma_s^{-1} = \lambda_{j(s)}$. Since $\lambda_j - \mu = \Delta(1 - \zeta_j)/2 > 0$ and $p^\star_N(\mu) = \epsilon^\star_N$, $K_s(\mu, \mu) = -\gamma_s p^\star_N(\mu)/(1 - \gamma_s \mu) = -2\epsilon^\star_N/(\Delta(1 - \zeta_{j(s)}))$, and $\sum_s K_s(\mu,\mu)^2 = 4(\epsilon^\star_N)^2 \Delta^{-2} \sum_j (1 - \zeta_j)^{-2}$ for every order. By (T2), $\sum_j (1 - \zeta_j)^{-2} = -\frac{d}{dz} \frac{T'_N}{T_N}\big|_{z=1} = T'_N(1)^2 - T''_N(1)$. The Chebyshev equation $(1 - z^2)T''_N - zT'_N + N^2 T_N = 0$ [18] and its derivative at $z = 1$ give $T'_N(1) = N^2$ and $T''_N(1) = N^2(N^2 - 1)/3$, hence the sum is $(2N^4 + N^2)/3 \geq 2N^4/3$, and A_N is at least the value at (μ, μ). □

B Proofs for Section 4

Theorem 4.1. By Corollary 7.1 (whose proof is independent of the present one), the method (1) is the Jacobi method with $P_s = 2^{1-s} T_s$ ($s \geq 1$), $P_0 = 1$, and every block starting at a site ≥ 1 has squared couplings $\frac{1}{4}$, so that $Q^{(s+1)}_{N-s-1} = 2^{s+1-N} U_{N-s-1}$. Theorem 5.2 with $P_N(\eta) = 2^{1-N} T_N(\eta)$ gives the kernel. By (T1), $|U_{N-s-1}| \leq N - s$ and $|T_s| \leq 1$ on $[-1, 1]$, with equality at $x = y = 1$. Hence the supremum is attained at $\lambda = \nu = \mu$, and $(A_N^{\mathrm{pref}})^2 = (\epsilon^\star_N/\Delta)^2(4N^2 + 16\sum_{m=1}^{N-1} m^2) = (\epsilon^\star_N/\Delta)^2(\frac{16}{3}N^3 + O(N^2))$. □

Uniqueness claim of Section 4. On $X = \mathbb{R}$, $H = \lambda$, exactness at $t = 1$ gives $1 - \gamma_0 \lambda = p^\star_1(\lambda) = 1 - 2\lambda/(L+\mu)$. For $t \geq 1$, exactness at $t-1, t, t+1$ gives $p^\star_{t+1} = (\alpha_t - \gamma_t \lambda)p^\star_t - \beta_t p^\star_{t-1}$, while the method (1) gives the same identity with the coefficients $1 + \beta^{\mathrm{GV}}_t$, $(1 + \beta^{\mathrm{GV}}_t)\frac{2}{L+\mu}$ and $\beta^{\mathrm{GV}}_t := T_{t-1}(\eta)/T_{t+1}(\eta)$. Subtracting yields a linear relation between $\lambda p^\star_t$, $p^\star_t$ and $p^\star_{t-1}$, which have degrees $t+1$, t, $t-1$; so all its coefficients vanish. □

C Proofs for Section 5

Theorem 5.1. Put $\bar\alpha_s := P_s(\eta)/P_{s+1}(\eta)$ and $\bar\beta_s := a_s^2 P_{s-1}(\eta)/P_{s+1}(\eta)$, so $\bar\beta_0 = 0$. Evaluating (2) at η gives $\bar\alpha_s(\eta - b_s) - \bar\beta_s = 1$, so with $e_s = x_s - x^\star$ the method reads $e_{s+1} = \bar\alpha_s(\eta - b_s - \frac{2}{\Delta}H)e_s - \bar\beta_s e_{s-1}$. Evaluating (2) at $z(\lambda) = \eta - 2\lambda/\Delta$ and dividing by $P_{s+1}(\eta)$ shows that $r_s(\lambda) = P_s(z(\lambda))/P_s(\eta)$ satisfies the same recurrence with H replaced by λ, $r_0 = 1$, $r_{-1} = 0$. By the spectral theorem $e_s = r_s(H)e_0$, and $r_N = p^\star_N$ if and only if $P_N = 2^{1-N} T_N$. □

Theorem 5.2. Apply Lemma A.1 with $\gamma_s = \frac{2}{\Delta} P_s(\eta)/P_{s+1}(\eta)$ and $r_s(\nu) = P_s(y)/P_s(\eta)$. In the variable $x = z(\lambda)$ the tail recurrence is $u_{t+1} = \frac{P_t(\eta)}{P_{t+1}(\eta)}(x - b_t)u_t - a_t^2 \frac{P_{t-1}(\eta)}{P_{t+1}(\eta)} u_{t-1}$. The sequence $u_t := \frac{P_{s+1}(\eta)}{P_t(\eta)} \det(xI - J_{s+1:t-1})$, $t \geq s+1$, with $u_s = 0$, satisfies it, because the determinants of the blocks $J_{s+1:t-1}$ obey (2) (expansion along the last row). Hence $G_{N,s+1} = \frac{P_{s+1}(\eta)}{P_N(\eta)} Q^{(s+1)}_{N-s-1}(x)$, and the product of the three factors is the stated kernel. The formula for $A_N(J)$ follows because z maps $[\mu, L]$ onto $[-1, 1]$. □

Finite Hessian drift. Let $D_{a:b}(x) := \det(xI - J_{a:b})$, $D_{a:a-1} := 1$, $D_{a:a-2} := 0$, so that $\widetilde{D}_{a:b} = 2^{b-a+1}D_{a:b}$, and, by expansion along the last row [32], the *block recurrence* $D_{a:t} = (x - b_t)D_{a:t-1} - a_t^2 D_{a:t-2}$, $t \geq a$, holds.

Theorem 7.1. (i) Let $x_s - x^\star = W_s(x_0 - x^\star)$ when the gradient at step s is $H_s(x_s - x^\star)$, $H_s = H + E_s$. Multiplying the update of Theorem 5.1 by $P_{s+1}(\eta)$ shows that $\Pi_s := P_s(\eta)W_s$ satisfies $\Pi_{-1} = 0$, $\Pi_0 = I$ and $\Pi_{s+1} = (Z_s - b_s)\Pi_s - a_s^2\Pi_{s-1}$, where $Z_s := Z - \frac{2}{\Delta}E_s$, and $\mathcal{T}_{\mathcal{R}_J}(H + E) = \Pi_N/P_N(\eta)$. We claim that

$$\Pi_t = D_{0:t-1}(Z) - \tfrac{2}{\Delta}\textstyle\sum_{r=0}^{t-1} D_{r+1:t-1}(Z)\, E_r\, \Pi_r, \qquad 0 \leq t \leq N. \tag{7}$$

It holds for $t = 0$ (and gives $\Pi_{-1} = 0$ for $t = -1$). If it holds for $t - 1$ and t, extend the sum for Π_{t-1} to $r = t - 1$ ($D_{t:t-2} = 0$), substitute both into the recurrence and use the block recurrence; this gives $\Pi_{t+1} = D_{0:t}(Z) - \frac{2}{\Delta}\sum_{r<t} D_{r+1:t}(Z)E_r\Pi_r - \frac{2}{\Delta}E_t\Pi_t$, which is the claim for $t + 1$. Substituting (7) into itself until no Π_r remains gives Π_N as the sum over $s_1 < \cdots < s_k$ of $(-\frac{2}{\Delta})^k D_{s_k+1:N-1}(Z)E_{s_k}\cdots E_{s_1}D_{0:s_1-1}(Z)$. The blocks have total length $N - k$, and $(-2/\Delta)^k 2^{-(N-k)}/P_N(\eta) = \frac{\epsilon_N^\star}{2}(-4/\Delta)^k$, which is (i); the term $k = 0$ is $\frac{\epsilon_N^\star}{2}\,2T_N(Z) = p_N^\star(H)$, and the term $k = 1$ is the linear part, $\mathcal{T}'_{\mathcal{R}_J,H}[E]$.

(ii) Z is symmetric with spectrum in $[-1, 1]$, so $\|\widetilde{D}_{a:b}(Z)\| \leq g_{a:b}$; submultiplicativity gives (ii).

(iii) $\det(\mathrm{diag}(1 + t_s) - J)$ is affine in each t_s, and the coefficient of $\prod_{s\in S} t_s$ is the principal minor of $I - J$ without the sites in S, which for a tridiagonal J is $D_{0:s_1-1}(1)D_{s_1+1:s_2-1}(1)\cdots D_{s_k+1:N-1}(1)$. Setting all $t_s = t$ and using $\det((1 + t)I - J) = 2^{1-N}T_N(1+t)$ gives $\sum_{s_1<\cdots<s_k} \widetilde{D}_{0:s_1-1}(1)\cdots\widetilde{D}_{s_k+1:N-1}(1) = 2^{1-k}[t^k]T_N(1+t)$. Since $g_{a:b} \geq |\widetilde{D}_{a:b}(1)|$, $G_k(J) \geq 2^{1-k}[t^k]T_N(1 + t)$, with equality when every block is maximal at 1. Then (i)–(ii) give $\|\mathcal{T} - p_N^\star(H)\| \leq \sum_{k\geq 1}\frac{\epsilon_N^\star}{2}(2u)^k 2^{1-k}[t^k]T_N(1 + t) = \epsilon_N^\star(T_N(1 + u) - 1)$; the term $k = 1$ gives $\|\mathcal{T}'\| \leq \epsilon_N^\star T_N'(1)u = \epsilon_N^\star N^2 u$, and the terms $k \geq 2$ give $\epsilon_N^\star(T_N(1 + u) - 1 - N^2u)$. Finally $\cosh\sqrt{2u} \geq 1 + u$ gives $T_N(1+u) \leq \cosh y$, $y = N\sqrt{2u}$, and $\cosh y - 1 - \frac{y^2}{2} = \sum_{j\geq 2}\frac{y^{2j}}{(2j)!} \leq \frac{y^4}{24}\cosh y$.

(iv) For $E_s = -\delta I$, $\mathcal{T}_{\mathcal{R}} = p_{\mathcal{R},N}(H - \delta I) = p_N^\star(H - \delta I)$, with linear part $-\delta(p_N^\star)'(H)$. On the μ-eigenspace, $z(\mu - \delta) = 1 + u$, so the three quantities equal $\epsilon_N^\star(T_N(1 + u) - 1)$, $\epsilon_N^\star N^2 u$ and their difference. □

Corollary 7.1(a). $P_0 = 1$ and $P_s = 2^{1-s}T_s$, $s \geq 1$, satisfy (2) with $b_s = 0$, $a_1^2 = \frac{1}{2}$, $a_s^2 = \frac{1}{4}$ for $s \geq 2$. In Theorem 5.1, $P_1(\eta) = \eta$ gives $x_1 = x_0 - \frac{2}{L+\mu}\nabla f(x_0)$, and for $s \geq 1$ the coefficients of x_s, x_{s-1}, $\nabla f(x_s)$ are $2\eta T_s(\eta)/T_{s+1}(\eta) = 1+\beta_s$, $-\beta_s$ and $-(1+\beta_s)\frac{2}{L+\mu}$, by $2\eta T_s = T_{s+1}+T_{s-1}$ and $\Delta\eta = L+\mu$; this is (1), and it proves its prefix exactness. Leading blocks give $\widetilde{D}_{0:b} = 2^{b+1}P_{b+1} = 2T_{b+1}$; a block starting at $a \geq 1$ has all squared couplings $\frac{1}{4}$, so $\widetilde{D}_{a:a+j-1}$ satisfies the recurrence of U_j, i.e. $\widetilde{D}_{a:b} = U_{b-a+1}$. By (T1) every block is maximal at 1, and Theorem 7.1(iii)–(iv) apply. □

D Proofs for Section 6

Throughout, $N \geq 2$ (for $N = 1$ all statements are immediate), $h := \pi/N$, $\theta_j = (2j - 1)h/2$, $\zeta_j = \cos\theta_j$, and $P_s = P_s^{(N)}$ are the polynomials of Lemma D.1; by the proof of Theorem 6.1 they are the leading polynomials of $J_N^{\sin}$.

Lemma D.1 (Explicit leading polynomials)**.** *Let $P_0 = 1$, $P_1 = x$, $P_{s+1} = xP_s - a_{s,N}^2 P_{s-1}$ for $1 \leq s \leq N - 1$, with $a_{s,N}^2$ as in Theorem 6.1, and, for $0 \leq k, s \leq N - 1$,*

$$r_{k,N} := \textstyle\prod_{j=1}^{k}\frac{\sin((j-\frac{1}{2})h)}{\sin(jh)} > 0, \qquad S_s(\theta) := \sum_{k=0}^{s} r_{k,N}r_{s-k,N}\cos((s - 2k)\theta). \tag{8}$$

Then $P_s(\cos\theta) = 2^{-s}r_{s,N}^{-1}S_s(\theta)$ for $0 \leq s \leq N - 1$, and $P_N = 2^{1-N}T_N$. In particular, $P_s(\cos\theta)/P_s(1) = \sum_{k=0}^{s} c_{s,k}\cos((s - 2k)\theta)$ with $c_{s,k} = r_{k,N}r_{s-k,N}/S_s(0) > 0$ and $\sum_k c_{s,k} = 1$.

These are the continuous q-ultraspherical (Rogers) polynomials $C_s(x; \beta \mid q)$ [14] at $q = \mathrm{e}^{2ih}$, $\beta = \mathrm{e}^{ih}$, a root-of-unity case studied in [29]; the proof below is self-contained.

Proof. Write $r_k := r_{k,N}$, $S_{-1} := 0$ and $z = \mathrm{e}^{i\theta}$, and let products containing an index outside $\{0,\dots,N-1\}$, such as r_{-1}, be zero; since $r_k r_{s-k}$ is symmetric in $k \leftrightarrow s-k$, $S_s = \sum_{k=0}^{s} r_k r_{s-k} z^{s-2k}$. For $0 \le s \le N-1$ the coefficient of z^{s+1-2k}, $0 \le k \le s+1$, in $2\cos\theta\,\sin((s+\frac12)h)\,S_s - \sin(sh)\,S_{s-1}$ is

$$c_k := \sin\big((s+\tfrac12)h\big)\,(r_k r_{s-k} + r_{k-1} r_{s+1-k}) - \sin(sh)\,r_{k-1} r_{s-k}.$$

For $1 \le k \le s$, put $A = kh$, $B = (s+1-k)h$, $u = h/2$, so that $s+\frac12 = (A+B-u)/h$, $s = (A+B-2u)/h$ and $r_k = r_{k-1}\sin(A-u)/\sin A$, $r_{s+1-k} = r_{s-k}\sin(B-u)/\sin B$. After division by $r_{k-1}r_{s-k} > 0$ and multiplication by $\sin A \sin B > 0$, the equality $c_k = \sin(A+B)\,r_k r_{s+1-k}$ becomes

$$\begin{aligned}&\sin(A+B-u)\big[\sin(A-u)\sin B + \sin(B-u)\sin A\big]\\ &\quad = \sin(A+B)\sin(A-u)\sin(B-u) + \sin(A+B-2u)\sin A\sin B,\end{aligned}$$

which holds for all A, B, u: by the product-to-sum formulas both sides equal $\frac14[\sin 2A + \sin 2B + \sin(2A-2u) + \sin(2B-2u)] - \frac12\sin(2A+2B-2u)$. For $k = 0$ and $k = s+1$, $c_k = \sin((s+\frac12)h)\,r_s$. Hence, if $s \le N-2$, so that $r_{s+1} = r_s \sin((s+\frac12)h)/\sin((s+1)h)$ is defined,

$$2\cos\theta\,\sin\big((s+\tfrac12)h\big)\,S_s = \sin\big((s+1)h\big)\,S_{s+1} + \sin(sh)\,S_{s-1}, \tag{9}$$

while for $s = N-1$, where $\sin(Nh) = 0$, only $k = 0, N$ contribute: $2\cos\theta\,\sin((N-\frac12)h)\,S_{N-1} - \sin((N-1)h)\,S_{N-2} = 2\sin((N-\frac12)h)\,r_{N-1}\cos N\theta$.

For $s \ge 1$ the terms $k = 0, s$ of S_s give $2r_s\cos s\theta = 2r_s T_s(x)$, $x = \cos\theta$, and the others have lower degree; with $S_0 = 1$, $\widetilde P_s(x) := 2^{-s} r_s^{-1} S_s(\theta)$ is a monic polynomial of degree s, with $\widetilde P_0 = 1$, $\widetilde P_1 = x$. Dividing (9) by $2^{s+1} r_s \sin((s+\frac12)h)$ and using $r_{s-1}/r_s = \sin(sh)/\sin((s-\frac12)h)$ gives $\widetilde P_{s+1} = x\widetilde P_s - a_{s,N}^2 \widetilde P_{s-1}$, so $\widetilde P_s = P_s$ for $s \le N-1$; the same division of the identity for $s = N-1$ gives $P_N = 2^{1-N} T_N$. Finally $P_s(1) = 2^{-s} r_s^{-1} S_s(0) > 0$. □

Theorem D.1 (Endpoint asymptotics of the sine hard-edge recurrence)**.** *Let $F_s = F_s^{(N)}$, $0 \le s \le N$, be defined by $F_0 = 1$, $F_1 = 2$ and $F_{s+1} - 2F_s + F_{s-1} = -\frac{\sin^2(h/2)}{\sin^2(sh) - \sin^2(h/2)} F_{s-1}$, $1 \le s < N$, and let f be as in Section 6. Then $f > 0$ is the unique solution of $f'' + \frac{\pi^2}{4\sin^2(\pi t)} f = 0$ with $f(t) \sim \sqrt{t}$ as $t \downarrow 0$; $F_{\lfloor Nt\rfloor}/\sqrt{\pi N} \to f(t)$ uniformly on compact subsets of $(0,1)$; $0 < F_s \le M\sqrt{N}$ for all $s \le N$, with $M = 3\pi^2/2$; and $G_N(t) := F_{\lfloor Nt\rfloor} F_{N-\lfloor Nt\rfloor - 1}/(\pi N)$ converges to $f(t)f(1-t)$ in $L^p(0,1)$, $1 \le p < \infty$.*

Proof. 1) Convolution. The monic recurrence of Lemma D.1 at $x = 1$, multiplied by 2^{s+1}, is the recurrence of the statement, because $\sin((s-\frac12)h)\sin((s+\frac12)h) = \sin^2(sh) - \sin^2(h/2)$; hence $F_s = 2^s P_s(1)$ for $0 \le s \le N$, and, by Lemma D.1 at $\theta = 0$, $F_s = r_{s,N}^{-1} \sum_{k=0}^{s} r_{k,N} r_{s-k,N}$ for $0 \le s < N$.

2) Asymptotics of $r_{k,N}$. Let $b_k := \binom{2k}{k} 4^{-k}$ and $g(x) := \log\frac{\sin \pi x}{\pi x}$. For fixed $0 < b < 1$, Taylor's formula on each mesh interval gives, uniformly for $k \le bN$, $\log(r_{k,N}/b_k) = \sum_{j\le k}[g(\frac{j-1/2}{N}) - g(\frac{j}{N})] = -\frac12 g(\frac{k}{N}) + O_b(N^{-1})$. With Stirling's formula [22, Eq. 5.11.3] $b_k\sqrt{\pi k} \to 1$, this gives $\sqrt{N}\, r_{\lfloor Nu\rfloor,N} \to (\sin\pi u)^{-1/2}$ locally uniformly on $(0,1)$, and $c_b(k+1)^{-1/2} \le r_{k,N} \le C_b(k+1)^{-1/2}$ for $k \le bN$.

3) Limit of the convolution. Let $S_{s,N} := \sum_k r_{k,N} r_{s-k,N}$ and $I(t) := \int_0^t (\sin\pi u \sin\pi(t-u))^{-1/2} du$. Uniformly for $s/N \in [a,b] \subset (0,1)$, $S_{s,N} \to I(s/N)$: on $\delta \le k/N \le s/N - \delta$ the sum is a Riemann sum with error $o(1)$ by Step 2 and uniform continuity, while the two omitted tails are at most $C_{a,b} N^{-1/2} \sum_{k\le \delta N} (k+1)^{-1/2} \le C'_{a,b}(\sqrt{\delta} + N^{-1/2})$, and the omitted integral tails at most $C''_{a,b}\sqrt{\delta}$. Hence $F_{\lfloor Nt\rfloor}/\sqrt{N} \to \sqrt{\sin\pi t}\, I(t)$ locally uniformly.

4) Elliptic form. With $k = \sin(\pi t/2)$, $y = \pi(u - t/2)$ and $\sin y = k\sin\theta$, $I(t) = \frac{1}{\pi}\int_{-\pi t/2}^{\pi t/2} (k^2 - \sin^2 y)^{-1/2} dy = \frac{2}{\pi}\mathrm{K}(k)$, which is the stated limit. By [22, Eq. 19.5.1], $H(z) := \frac{2}{\pi}\mathrm{K}(\sqrt{z}) = \sum_j b_j^2 z^j = {}_2F_1(\frac12, \frac12; 1; z)$, which satisfies the hypergeometric equation [22, Eq. 15.10.1] $z(1-z)H'' + (1-2z)H' - H/4 = 0$; for $Y(T) = H(\sin^2\frac{T}{2})$ this gives $Y'' + \cot T\, Y' - \frac14 Y = 0$, and $\sqrt{\sin T}\, Y$ solves $w'' + w/(4\sin^2 T) = 0$; rescaling $T = \pi t$ gives the equation for f. Since $H(0) = 1$, $f(t) \sim \sqrt{t}$, and the integral shows $f > 0$. The second solution $f\int_{t_0}^t f^{-2}$ behaves like $\sqrt{t}\log t$, so the condition $f \sim \sqrt{t}$ selects f.

5) Uniform bound. g is decreasing, and for $k \le N/2$ the half-mesh intervals in $\log(r_{k,N}/b_k) = \sum_{j\le k}\int_{(j-1/2)/N}^{j/N} -g'$ are disjoint subsets of $[0,k/N]$, so $b_k \le r_{k,N} \le \frac{\pi}{2} b_k$. From $\sum_k b_k z^k = (1-z)^{-1/2}$, $\sum_{k\le s} b_k b_{s-k} = 1$, and $b_s \ge 1/(2\sqrt{s})$ for $s \ge 1$, Step 1 and $F_0 = 1$ give $0 < F_s \le \frac{\pi^2}{2}\sqrt{s+1}$ for $s \le m := \lfloor N/2 \rfloor$. Also $F_N = 2^N P_N(1) = 2 > 0$, and the recurrence has negative second differences, so $s \mapsto F_s$ is concave on $[0,N]$. Concavity at m gives $F_s \le \frac{N-s}{N-m} F_m$ for $s \le m$ and $F_s \le \frac{s}{m} F_m$ for $s \ge m$, both factors at most 3, hence $F_s \le \frac{3\pi^2}{2}\sqrt{N}$.

6) Steps 3–4 give $G_N(t) \to f(t)f(1-t)$ for every $t \in (0,1)$, and Step 5 gives $0 \le G_N \le M^2/\pi$; dominated convergence proves the L^p convergence. □

Theorem 6.1. Let J be the Jacobi matrix with zero diagonal and off-diagonal entries $a_{s,N} > 0$; its leading polynomials are the P_s of Lemma D.1, so $\det(xI - J) = P_N = 2^{1-N} T_N$ and $\operatorname{spec}(J) = \{\zeta_j\}$. The formula for $a_{s,N}^2$ is invariant under $s \mapsto N - s$, so J is persymmetric, and by the criterion of Section 2.3 its weights are proportional to $1/|P_N'(\zeta_j)| = 2^{N-1} \sin\theta_j / N$ (T4). These are the sine weights, and a Jacobi matrix is determined by its spectral measure (Section 2.3), so $J = J_N^{\sin}$. The asymptotics of $a_{s,N}^2$ follow from $a_{s,N}^2 = \frac{1}{4}(1 - \sin^2(h/2)/\sin^2(\pi t))^{-1}$, $t = s/N$. By Lemma D.1, $|P_s(\cos\theta)|/P_s(1) \le \sum_k c_{s,k} = 1$; the principal blocks are treated in Lemma D.2. □

Theorem 6.2. By persymmetry, the tail block on $s+1, \dots, N-1$ is the leading block of length $N-s-1$ reversed, so $Q_{N-s-1}^{(s+1)} = P_{N-s-1}$. By (3) and $P_N(1) = 2^{1-N}$, $d_s = P_s(1) P_{N-s-1}(1)/P_N(1) = F_s F_{N-s-1}$, where $F_s = 2^s P_s(1)$ is the sequence of Theorem D.1 (Step 1 of its proof). Thus $\varphi_N = \pi G_N \to \pi f(t) f(1-t)$ in L^2. By Lemma 5.1, $\int_0^1 \varphi_N = N^{-2} \sum_s d_s = 1$, so $\int_0^1 f(t) f(1-t)\, dt = 1/\pi$, the limit is $\varphi_{\sin}$, and $N^{-3} \sum_s d_s^2 = \int_0^1 \varphi_N^2 \to c_{\sin}$. Exactness follows from $P_N = 2^{1-N} T_N$ and Theorem 5.1. Both P_s and $Q_{N-s-1}^{(s+1)} = P_{N-s-1}$ satisfy the maximum principle, so the supremum in Theorem 5.2 is attained at $x = y = 1$, and $A_N = \frac{2}{\Delta P_N(\eta)} (\sum_s P_s(1)^2 P_{N-s-1}(1)^2)^{1/2} = \frac{2P_N(1)}{\Delta P_N(\eta)} (\sum_s d_s^2)^{1/2}$ with $P_N(1)/P_N(\eta) = \epsilon_N^\star$. □

Lemma D.2 (Fourier positivity of the sine blocks)**.** *Let $J = J_N^{\sin}$, $D_{a:b}$ as in Appendix C (so $D_{0:s-1} = P_s$), and $D_{a:b}(\cos\theta) = \sum_{|j|\le b-a+1} d_j \mathrm{e}^{ij\theta}$. Then all $d_j \ge 0$; consequently $|D_{a:b}(x)| \le D_{a:b}(1)$ on $[-1,1]$.*

Proof. The consequence follows from $|D_{a:b}(\cos\theta)| \le \sum_j d_j$. We use only Lemma D.1 and persymmetry, established in the proof of Theorem 6.1 before the blocks; thus $D_{a:N-1} = P_{N-a}$. *End blocks.* $D_{0:N-1} = 2^{1-N} T_N$; for $b \le N-2$, $D_{0:b} = P_{b+1}$ is covered by Lemma D.1, and so is $D_{a:N-1} = P_{N-a}$ for $a \ge 1$. *Interior blocks.* Let $1 \le a \le b \le N-2$, $\ell := b - a + 1$. We first show the continuant identity

$$P_N\, D_{a:b} = P_{b+1}\, D_{a:N-1} - \Big(\prod_{k=a}^{b+1} a_k^2\Big) D_{0:a-2}\, D_{b+2:N-1}, \tag{10}$$

where a_k are the off-diagonal entries of J. For $a-2 \le m \le N-2$ let $F(m) := P_{m+1} D_{a:N-1} - P_N D_{a:m}$ and $G(m) := (\prod_{k=a}^{m+1} a_k^2) D_{0:a-2} D_{m+2:N-1}$. By the block recurrence, $F(m+1) = xF(m) - a_{m+1}^2 F(m-1)$; expanding $D_{m+1:N-1}$ along its first row, $a_{m+2}^2 D_{m+3:N-1} = x D_{m+2:N-1} - D_{m+1:N-1}$, and G satisfies the same recurrence. At $m = a-2$, $F = G = P_{a-1} D_{a:N-1}$; at $m = a-1$, splitting $P_N = P_a D_{a:N-1} - a_a^2 P_{a-1} D_{a+1:N-1}$ gives $F = G$. Hence $F(b) = G(b)$. With $x = \cos\theta$, $z = \mathrm{e}^{i\theta}$, the left side of (10) is $2^{-N}(z^N + z^{-N}) \sum_{|j|\le \ell} d_j z^j$, whose exponents lie in the disjoint sets $\pm N + [-\ell, \ell]$. The last term has degree $N - \ell - 2$ in x. Hence $d_j = 2^N [z^{N+j}](P_{b+1} P_{N-a})(\cos\theta)$ for $|j| \le \ell$, and both factors, hence their product, have nonnegative coefficients by Lemma D.1, since $b+1, N-a \le N-1$. □

Corollary 7.1(b). $\widetilde{D}_{a:b}$ is a positive multiple of $D_{a:b}$, so by Lemma D.2 every block is maximal at 1, and Theorem 7.1(iii)–(iv) apply. □

E Proofs for Section 7

Theorem 7.2. Let J be the Jacobi matrix of the method (Corollary 7.1 or Theorem 6.1), $H := \nabla^2 f(x^\star)$, $Z := z(H)$, $e_s := x_s - x^\star$, $r_0 := \|e_0\| \le r$. Since $\nabla f(x^\star) = 0$, $\nabla f(x_s) = \bar{H}_s e_s$ with $\bar{H}_s = \int_0^1 \nabla^2 f(x^\star + \tau e_s)\, d\tau$, as long as $x_s \in \mathbb{B}$; put $E_s := \bar{H}_s - H$. Then $\|E_s\| \le \int_0^1 \rho\tau \|e_s\|\, d\tau = \rho\|e_s\|/2$.

Expansion at intermediate times. As in the proof of Theorem 7.1(i), $\Pi_t := P_t(\eta)W_t$, where $e_t = W_t e_0$, satisfies the Duhamel formula (7) for every $0 \le t \le N$; iterating it as in Theorem 7.1(i) expresses Π_t through $E_0, \dots, E_{t-1}$ and blocks of $J_{0:t-1}$. Every block $D_{a:b} = \det(xI - J_{a:b})$ here (notation of Appendix C) is a block of J; by Corollary 7.1, $|D_{a:b}(x)| \le D_{a:b}(1)$ on $[-1,1]$, and $D_{a:b}(1) > 0$ because the eigenvalues of $J_{a:b}$ interlace those of J [30] and so lie in $(-1,1)$. If $\|E_s\| \le \delta$ for $s < t$, then, with $u := 2\delta/\Delta$, $\|\Pi_t\| \le \sum_k u^k \sum_{s_1<\cdots<s_k<t} D_{0:s_1-1}(1)\cdots D_{s_k+1:t-1}(1) = \det((1+u)I - J_{0:t-1}) = P_t(1+u)$, by the principal-minor expansion of $\det(\mathrm{diag}(1+t_s) - J_{0:t-1})$ used in the proof of Theorem 7.1(iii). Hence $\|W_t\| \le P_t(1+u)/P_t(\eta)$. The zeros of P_t lie in $(-1,1)$, so P_t is positive and increasing on $[1,\infty)$; if $\delta \le \mu$ then $1 + u \le 1 + 2\mu/\Delta = \eta$ and $\|W_t\| \le 1$.

Induction. Put $\delta := \rho r/2 \le \mu$. We show $\|e_t\| \le r_0$ for $0 \le t \le N$. This holds for $t = 0$. If it holds for all $s < t$, then $x_s \in \mathbb{B}$ and $\|E_s\| \le \rho r_0/2 \le \delta$ for $s < t$, so $\|e_t\| \le \|W_t\| r_0 \le r_0$. For $t = N$ the same estimate gives $\|e_N\| \le \frac{P_N(1+u)}{P_N(\eta)}\, r_0 = \epsilon_N^\star\, T_N\Big(1 + \frac{\rho r}{\Delta}\Big)\, r_0$.

Explicit rate. Write $\eta = \cosh a$ and $1+u = \cosh b$ with $0 \le b \le a$. Then $\epsilon_N^\star T_N(1+u) = \frac{\cosh(Nb)}{\cosh(Na)} \le 2e^{-N(a-b)}$. Since $\operatorname{arccosh}(1+2v) = 2\operatorname{arcsinh}\sqrt{v}$, $a = 2\operatorname{arcsinh}\sqrt{\mu/\Delta}$, $\quad b = 2\operatorname{arcsinh}\sqrt{\delta/\Delta}$. The derivative of arcsinh is $(1+v^2)^{-1/2} \ge \sqrt{\Delta/L}$ for $0 \le v \le \sqrt{\mu/\Delta}$, so $a - b \ge \frac{2(\sqrt{\mu}-\sqrt{\delta})}{\sqrt{L}} = \frac{2}{\sqrt{\kappa}}\Big(1 - \sqrt{\delta/\mu}\Big)$. This proves the displayed bound.

Restarts. If $\rho r_0 \le \mu/2$, apply the above with $r = r_0$. Then $\sqrt{\rho r_0/(2\mu)} \le \frac{1}{2}$, so $\|e_N\| \le 2e^{-N/\sqrt{\kappa}} r_0$, which is at most $r_0/2$ for $N \ge \sqrt{\kappa}\ln 4$. The output is again in the region $\rho\|x - x^\star\| \le \mu/2$, so the argument repeats, and $\lceil \log_2(r_0/\varepsilon) \rceil$ blocks suffice. □

Theorem 7.3. Keep the notation of the proof of Theorem 7.2, now with $\delta := \rho r_0/2$ and $u := 2\delta/\Delta$. The Duhamel formula (7) at $t = N$, applied to e_0, together with $D_{0:N-1}(Z) = P_N(Z)$, $\Pi_r e_0 = P_r(\eta) e_r$ and $P_N(Z)/P_N(\eta) = p_N^\star(H)$, gives the exact identity

$$e_N - p_N^\star(H) e_0 = -\tfrac{2}{\Delta P_N(\eta)} \textstyle\sum_{r=0}^{N-1} P_r(\eta)\, D_{r+1:N-1}(Z)\, E_r e_r. \tag{11}$$

Here $\|E_r e_r\| \le \frac{\rho}{2}\|e_r\|^2$, $\|D_{r+1:N-1}(Z)\| \le D_{r+1:N-1}(1) = Q_{N-r-1}^{(r+1)}(1)$, and, by the proof of Theorem 7.2, $\|e_r\| \le r_0 P_r(1+u)/P_r(\eta)$. Hence

$$\|e_N - p_N^\star(H) e_0\| \le \tfrac{\rho r_0^2}{\Delta P_N(\eta)} \textstyle\sum_r Q_{N-r-1}^{(r+1)}(1)\, \frac{P_r(1+u)^2}{P_r(\eta)} = \frac{\rho r_0^2}{\Delta}\, \epsilon_N^\star \sum_r d_r\, m_r \Big(\frac{P_r(1+u)}{P_r(1)}\Big)^2,$$

because $Q_{N-r-1}^{(r+1)}(1)\, P_r(1) = d_r P_N(1)$, $P_N(1) = \epsilon_N^\star P_N(\eta)$ and $P_r(1) = m_r P_r(\eta)$. Let $\zeta_i^{(r)} \in (-1,1)$ be the zeros of P_r and $A := I - J \succ 0$. Then $\frac{P_r(1+u)}{P_r(1)} = \prod_i \Big(1 + \frac{u}{1-\zeta_i^{(r)}}\Big) \le \exp\big(u\,\mathrm{tr}(A_{0:r-1})^{-1}\big)$. For a positive definite A, the inverse of a principal block is dominated by the corresponding block of the inverse, $(A_{0:r-1})^{-1} \preceq (A^{-1})_{0:r-1}$, because the latter is the inverse of a Schur complement $A_{0:r-1} - A_{12}A_{22}^{-1}A_{21} \preceq A_{0:r-1}$ [9]. Hence $\mathrm{tr}(A_{0:r-1})^{-1} \le \mathrm{tr}\, A^{-1} = N^2$ by Lemma 5.1, and the bound follows.

For the example, $H = \mu$, $Z = 1$ and $E_r = -\frac{\rho}{2} e_r$, so (11) becomes a sum of nonnegative terms, $e_N - p_N^\star(\mu) e_0 = \frac{\rho}{\Delta P_N(\eta)} \sum_r P_r(\eta)\, Q_{N-r-1}^{(r+1)}(1)\, e_r^2$. Since $e_r = r_r(\mu) e_0 + O(r_0^2) = m_r r_0 + O(r_0^2)$ and $P_r(\eta) m_r^2 = P_r(1) m_r$, the right-hand side equals $\frac{\rho}{\Delta}\epsilon_N^\star C_N(J)\, r_0^2\, (1 + O(r_0))$. □

F Proofs for Section 8

Proposition 8.2. By Theorem 5.2 with $s = 0$ and $P_0 = 1$, $K_0(x,y) = -\frac{2}{\Delta P_N(\eta)} Q_{N-1}^{(1)}(x)$ does not depend on y, and at $x = 1$ it equals $-\frac{2\epsilon_N^\star}{\Delta} d_0$ by (5). For the semi-iterative method $d_0 = N$ (Section 5). In general $d_0 = \langle (I-J)^{-1}\mathbf{1}_0, \mathbf{1}_0\rangle = \sum_j w_j/(1-\zeta_j)$ by the spectral theorem, w being the spectral measure of the first basis vector. For the sine weights $w_j \propto \sin\theta_j$, $\sin\theta/(1-\cos\theta) = \cot(\theta/2)$ and $\sum_k \sin\theta_k = \sin^2(N\cdot\frac{\pi}{2N})/\sin\frac{\pi}{2N} = 1/\sin\frac{\pi}{2N}$, which gives the closed form.

For the asymptotics write $\cot\frac{\theta}{2} = \frac{2}{\theta} + \phi(\theta)$, where ϕ is smooth on $[0,\pi]$. The points θ_j are the midpoints of a uniform partition of $[0,\pi]$ with step $h = \pi/N$, so the midpoint rule gives $\sum_j \phi(\theta_j) = h^{-1}\int_0^\pi \phi + O(h)$, and $\int_0^\pi \phi = \left[2\ln\sin\frac{\theta}{2} - 2\ln\theta\right]_0^\pi = 2\ln\frac{2}{\pi}$. Further, $\sum_j \frac{2}{\theta_j} = \frac{4N}{\pi}\sum_{j=1}^{N}\frac{1}{2j-1} = \frac{4N}{\pi}(H_{2N} - \frac{1}{2}H_N) = \frac{4N}{\pi}(\frac{1}{2}\ln N + \ln 2 + \frac{1}{2}\gamma_{\mathrm{E}} + O(N^{-2}))$, the terms of order $1/N$ in the expansions of the harmonic numbers [22, Eq. 5.4.14, Eq. 5.11.2] cancelling. Hence $\sum_j \cot\frac{\theta_j}{2} = \frac{2N}{\pi}(\ln N + \gamma_{\mathrm{E}} + \ln\frac{8}{\pi}) + O(\frac{\ln N}{N})$, and multiplication by $\sin\frac{\pi}{2N} = \frac{\pi}{2N}(1+O(N^{-2}))$ proves the expansion. Persymmetry gives $R(I-J)^{-1}R = (I-J)^{-1}$, so $d_{N-1} = d_0$. □

Corollary 8.1. (a) Here $\xi_s = 0$ and $\eta_s = \nabla f(x_s) - He_s = \int_0^1(\nabla^2 f(x^\star + \tau e_s) - H)e_s\,d\tau = \frac{1}{2}\nabla^3 f(x^\star)[e_s,e_s] + o(\|e_s\|^2)$, in particular $\eta_s = O(\|e_s\|^2)$. Lemma 8.1 gives $e_N - p_N(H)e_0 = -\sum_s \gamma_s G_s \eta_s$ exactly, and, applied at time s by induction, $e_s = r_s(H)e_0 + O(\|e_0\|^2)$. Substitution gives $Z(e_0)$ up to $o(\|e_0\|^2)$. With $e_0 = r_0 v$, $Z(e_0) = r_0^2 Z(v)$, and the bound on $\|Z(v)\|$ follows from Theorem 7.3, divided by r_0^2, as $r_0 \to 0$.

(b) Since $\mathcal{L}_s\mathbf{1} = 0$ and $\alpha_s - \beta_s = 1$, $\mathbf{1}^\top x_{s+1} = \alpha_s\mathbf{1}^\top x_s - \beta_s\mathbf{1}^\top x_{s-1}$ and $x_{-1} = x_0$, so the average is preserved. With the consensus target $x^\star = \bar{x}_0\mathbf{1}$, we have $e_s \perp \mathbf{1}$. On $\mathbf{1}^\perp$ the spectrum of $\mathcal{L}$ lies in $[\lambda_2, \lambda_{\max}]$, and $b_e \perp \mathbf{1}$. With independent indicators χ_e of mean $1-p$, $\eta_s = \sum_e(\chi_e/(1-p) - 1)w_e b_e(b_e^\top e_s)$, and $\mathrm{Var}(\chi_e/(1-p)) = p/(1-p)$ gives the conditional covariance $\frac{p}{1-p}\sum_e w_e^2(b_e^\top e_s)^2\, b_e b_e^\top$, which is positive semidefinite and quadratic in e_s. Apply Proposition 8.1 with $\sigma^2 = p/(1-p)$. Without the rescaling, the noise variance is $p(1-p)$ and every γ_s is divided by $1-p$, which gives the same product. □

Supplementary Material

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Notation and unprefixed references are those of *Chebyshev-Exact Acceleration under Hessian Variation, I: Sine–Jacobi Method*.

S1 Persymmetric Jacobi matrices

Lemma S1.1. *Let J be an $N \times N$ Jacobi matrix as in Section 2.3, with positive off-diagonal entries $a_1, \dots, a_{N-1}$, distinct eigenvalues $\lambda_1 < \cdots < \lambda_N$, characteristic polynomial $P_N(x) = \prod_j (x - \lambda_j)$, and first spectral weights w_j. With $Re_s = e_{N-1-s}$,*

$$RJR = J \quad \Longleftrightarrow \quad w_j = \frac{c}{|P_N'(\lambda_j)|}, \qquad c = \left(\sum_j \frac{1}{|P_N'(\lambda_j)|} \right)^{-1}. \tag{S1}$$

Proof. For a normalized real eigenvector v_j, write $w_j = (v_j)_0^2$ and $\widetilde{w}_j = (v_j)_{N-1}^2$. Both endpoint coordinates are nonzero: the eigenvector recurrence would otherwise force every coordinate to vanish. Thus both sets of weights are positive and sum to one.

Put $A := a_1 \cdots a_{N-1}$, with the empty product equal to one. The cofactor formula and spectral decomposition give

$$\left[(zI - J)^{-1} \right]_{0,N-1} = \frac{A}{P_N(z)} = \sum_j \frac{(v_j)_0 (v_j)_{N-1}}{z - \lambda_j}.$$

Taking residues and squaring yields

$$w_j \widetilde{w}_j = \frac{A^2}{P_N'(\lambda_j)^2}. \tag{S2}$$

If $w_j = c/|P_N'(\lambda_j)|$, then $\widetilde{w}_j = (A^2/c)/|P_N'(\lambda_j)|$. Normalization of the two weight vectors implies $A^2/c = c$, and hence $\widetilde{w}_j = w_j$. The first spectral measure of RJR is the last spectral measure of J, so J and RJR have the same first measure.

To see that this measure determines the matrix, equip functions on its N support points with the inner product $\langle p, q \rangle = \sum_j w_j p(\lambda_j) q(\lambda_j)$. Gram–Schmidt applied to $1, x, \dots, x^{N-1}$ gives unique orthonormal polynomials with positive leading coefficients. Multiplication by x on these support points has a unique Jacobi matrix in that basis. Consequently $J = RJR$.

Conversely, if $RJR = J$, the first and last measures coincide. Equation (S2) then gives $w_j = A/|P_N'(\lambda_j)|$; normalization identifies $A = c$. The same argument includes $N = 1$. □

For the Chebyshev nodes, reindexed as $\lambda_j = \cos\theta_j$, $\theta_j = (2j-1)\pi/(2N)$, one has $|P_N'(\lambda_j)| = 2^{1-N} N / \sin\theta_j$; thus (S1) gives exactly the sine weights of Section 6.

S2 Finite-horizon gain comparison

Corollary S2.1. *For every $N \geq 2$, every final-Chebyshev-exact stepsize schedule has strictly larger gain than the semi-iterative method. More precisely,*

$$A_N(\mathcal{R})^2 - (A_N^{\mathrm{pref}})^2 \geq \frac{8(\epsilon_N^\star)^2}{3\Delta^2} N(N-1)(N^2 - N + 1) > 0. \tag{S3}$$

As $N \to \infty$, uniformly over all orders of the steps, $A_N(\mathcal{R})/A_N^{\mathrm{pref}} \geq \sqrt{N/2}\,(1 + o(1))$.

Proof. The proof of Theorem 3.1 gives the finite bound

$$A_N(\mathcal{R})^2 \geq \frac{4(\epsilon_N^\star)^2}{\Delta^2}\,\frac{2N^4+N^2}{3}.$$

The kernels of Theorem 4.1, evaluated at their common maximizer $\lambda = \nu = \mu$, give

$$(A_N^{\mathrm{pref}})^2 = \frac{4(\epsilon_N^\star)^2}{\Delta^2}\left(N^2 + 4\sum_{m=1}^{N-1} m^2\right) = \frac{4(\epsilon_N^\star)^2}{\Delta^2}\left(N^2 + \frac{2}{3}N(N-1)(2N-1)\right).$$

Subtracting proves (S3). The ratio of the two parenthetical expressions is $N/2\,(1+O(N^{-1}))$, which proves the asymptotic bound. At $N=1$ the two methods coincide. □

S3 Drift and target accuracy

For either momentum method, Corollary 7.1 gives the attained worst-case terminal factor

$$Q_N(\delta) := \epsilon_N^\star T_N\left(1+\frac{2\delta}{\Delta}\right) = \frac{\cosh(2N\,\mathrm{arsinh}\,\sqrt{\delta/\Delta})}{\cosh(2N\,\mathrm{arsinh}\,\sqrt{\mu/\Delta})}, \qquad \delta \geq 0. \tag{S4}$$

Its amplification relative to the unperturbed factor is $M_N(\delta) := Q_N(\delta)/\epsilon_N^\star$. For any prescribed $M > 1$, monotonicity of cosh on $[0,\infty)$ gives the exact equivalence

$$M_N(\delta) \leq M \quad\Longleftrightarrow\quad \delta \leq \delta_M(N) := \Delta\sinh^2\left(\frac{\operatorname{arccosh} M}{2N}\right). \tag{S5}$$

In particular, $\delta_M(N) \sim \Delta(\operatorname{arccosh} M)^2/(4N^2)$ as $N\to\infty$.

Let $0<\varepsilon<1$ and $\kappa = L/\mu$. The smallest horizon with $\epsilon_N^\star \leq \varepsilon$ is

$$N_\varepsilon = \left\lceil \frac{\operatorname{arccosh}(1/\varepsilon)}{2\,\mathrm{arsinh}(1/\sqrt{\kappa-1})} \right\rceil. \tag{S6}$$

For fixed ε, as $\kappa\to\infty$,

$$\frac{N_\varepsilon}{\sqrt{\kappa}} \longrightarrow \frac{1}{2}\operatorname{arccosh}(1/\varepsilon), \qquad \frac{\delta_M(N_\varepsilon)}{\mu} \longrightarrow \frac{(\operatorname{arccosh} M)^2}{(\operatorname{arccosh}(1/\varepsilon))^2}.$$

Since $\operatorname{arccosh}(1/\varepsilon) \sim \log(2/\varepsilon)$ as $\varepsilon\downarrow 0$, the scale for a fixed amplification is $\mu/\log^2(2/\varepsilon)$. Equivalently, in these successive limits, $N_\varepsilon \approx \frac{1}{2}\sqrt{\kappa}\log(2/\varepsilon)$ and $\Delta/N_\varepsilon^2 \approx 4\mu/\log^2(2/\varepsilon)$.

For a fixed fraction $q = \delta/\mu \in (0,1]$,

$$M_{N_\varepsilon}(q\mu) \longrightarrow \cosh\bigl(\sqrt{q}\,\operatorname{arccosh}(1/\varepsilon)\bigr) \quad (\kappa\to\infty).$$

The last expression is asymptotic to $\frac{1}{2}(2/\varepsilon)^{\sqrt{q}}$ as $\varepsilon\downarrow 0$. At $q=0$, the amplification is exactly one.

For $\kappa = 10^4$ and $N = 725$, exact rational evaluation of the Chebyshev recurrence gives $1.008\cdot 10^{-6} < \epsilon_{725}^\star < 1.009\cdot 10^{-6}$ and

δ/μ	δ/L	$M_{725}(\delta)$
0.01	10^{-6}	$2.24 < M_{725} < 2.25$
0.05	$5\cdot 10^{-6}$	$12.8 < M_{725} < 12.9$
0.25	$2.5\cdot 10^{-5}$	$704 < M_{725} < 705$

Here the evaluation points are $\eta = 10001/9999$ and $1+2q/9999$; the bounds follow using $T_0 = 1$, $T_1 = x$, $T_{n+1} = 2xT_n - T_{n-1}$ over $\mathbb{Q}$. The static drift $E_s = -\delta I$ attains the factor in (S4) on the μ-eigenspace (Theorem 7.1(iv)). Thus a drift negligible relative to L can still amplify the terminal error substantially; shortening the horizon increases the budget (S5).

S4 Random and nonlinear perturbations

Proposition S4.1. *Fix a length-N realization, a reference Hessian H, a deterministic initial error e_0, and symmetric operators $B_0, \dots, B_{N-1}$. Let z_s be independent real random variables with $\mathbb{E}z_s = 0$, $\mathbb{E}z_s^2 = 1$, and use the Hessians $H_s = H + \sigma z_s B_s$. There are deterministic vectors v_S, indexed by subsets of $\{0, \dots, N-1\}$, such that*

$$e_N(\sigma) = \sum_S \sigma^{|S|} z_S v_S, \qquad z_S := \prod_{s \in S} z_s, \qquad v_\varnothing = p_{\mathcal{R},N}(H) e_0. \tag{S7}$$

Consequently,

$$\mathbb{E}e_N(\sigma) = v_\varnothing, \qquad \mathbb{E}\|e_N(\sigma) - v_\varnothing\|^2 = \sum_{k=1}^{N} c_k \sigma^{2k}, \qquad c_k := \sum_{|S|=k} \|v_S\|^2 \geq 0. \tag{S8}$$

The leading coefficient is $c_1 = \sum_s \|\mathcal{T}_s'(B_s)e_0\|^2$, with $\mathcal{T}_s'$ as in Lemma 3.1.

Proof. Each factor of the ordered output operator (4) is affine in σz_s. Expanding the product gives (S7): the vector v_S is obtained by replacing precisely the factors indexed by S with their perturbation terms, omitting the scalars σz_s. In particular $v_{\{s\}} = \mathcal{T}_s'(B_s)e_0$.

For distinct subsets S, T, some index belongs to exactly one of them. Independence and centering therefore give $\mathbb{E}(z_S z_T) = 0$. For $S = T$, independence and unit variances give $\mathbb{E}z_S^2 = 1$. Every nonempty S also satisfies $\mathbb{E}z_S = 0$. Taking the mean and expanding the squared norm proves (S8). Only second moments are needed, since each variable has exponent at most two in any such product. □

In this model the remainder of Proposition 8.1 is exactly $\sum_{k=2}^{N} c_k \sigma^{2k}$. Replacing σ by $\sigma/\sqrt{b}$ gives a polynomial in σ^2/b with nonnegative coefficients. Higher insertion orders therefore cannot cancel the first-order contribution in expectation.

Proposition S4.2. *Let f be C^3 near its minimizer $x^\star$, with $H_\star = \nabla^2 f(x^\star)$ and $\mu I \preceq H_\star \preceq LI$. Fix a method and horizon as in Corollary 8.1(a), and start from $x_0 = x^\star + ru$, $\|u\| = 1$. Write*

$$a(u) := p_N(H_\star)u, \qquad b_{\mathcal{R}}(u) := Z_{\mathcal{R}}(u),$$

where $Z_{\mathcal{R}}$ is the quadratic term in that corollary. Extend $F_{\mathcal{R}}(r,u) := \|x_N - x^\star\|/r$, $r > 0$, continuously to $F_{\mathcal{R}}(0,u) := \|a(u)\|$. Then

$$\partial_r^+ F_{\mathcal{R}}(0,u) = \begin{cases} \dfrac{\langle a(u), b_{\mathcal{R}}(u)\rangle}{\|a(u)\|}, & a(u) \neq 0, \\ \|b_{\mathcal{R}}(u)\|, & a(u) = 0. \end{cases} \tag{S9}$$

Proof. By Corollary 8.1(a) and homogeneity of its quadratic term,

$$x_N - x^\star = r\,a(u) + r^2 b_{\mathcal{R}}(u) + o(r^2).$$

Thus $F_{\mathcal{R}}(r,u) = \|a(u) + r b_{\mathcal{R}}(u) + o(r)\|$. At a nonzero a, the derivative of the norm in direction b is $\langle a, b\rangle/\|a\|$. At $a = 0$, the last display equals $r\|b_{\mathcal{R}}(u)\| + o(r)$ for $r \downarrow 0$. This proves both cases. □

Since $a(-u) = -a(u)$ and $b_{\mathcal{R}}(-u) = b_{\mathcal{R}}(u)$, the nonzero slopes in (S9) have opposite signs in opposite directions when $a(u) \neq 0$. Hence the normalized error need not increase with the starting radius. A bound on the size of the quadratic term does not determine the sign of its inner product with the linear term.

S5 Algorithms

Algorithms S1 and S2 implement (1) and (6) using only ratios of adjacent normalizing polynomials. In Algorithm S1, the variable u at step s is $T_{s-1}(\eta)/T_s(\eta)$. The Chebyshev recurrence gives

$$\frac{T_{s+1}(\eta)}{T_s(\eta)} = 2\eta - u, \qquad \beta_s = \frac{u}{2\eta - u}, \qquad \gamma_s = \frac{4}{\Delta(2\eta - u)}.$$

In Algorithm S2, $u = p_{s-1}/p_s$, where $p_s = P_s(\eta)$; the Jacobi recurrence gives

$$\frac{p_{s+1}}{p_s} = \eta - a_{s,N}^2 u, \qquad \beta_s = \frac{a_{s,N}^2 u}{\eta - a_{s,N}^2 u}, \qquad \gamma_s = \frac{2}{\Delta(\eta - a_{s,N}^2 u)}.$$

All denominators are positive by $T_s(\eta) > 0$ and $P_s(\eta) > 0$ (Theorem 5.1). Each algorithm uses one gradient evaluation per step, N in total, and stores two iterates.

Algorithm S1 Chebyshev semi-iterative method

Require: x_0, gradient oracle ∇f, $0 < \mu < L$, integer $N \geq 1$

$\Delta \leftarrow L - \mu$, $\eta \leftarrow (L + \mu)/\Delta$, $u \leftarrow 1/\eta$
$x_1 \leftarrow x_0 - \frac{2}{L+\mu}\nabla f(x_0)$
for $s = 1, \ldots, N-1$ **do**
    $d \leftarrow 2\eta - u$, $\beta \leftarrow u/d$, $\gamma \leftarrow 4/(\Delta d)$
    $x_{s+1} \leftarrow x_s + \beta(x_s - x_{s-1}) - \gamma \nabla f(x_s)$
    $u \leftarrow 1/d$
end for
return x_N

Algorithm S2 Sine–Jacobi method

Require: x_0, gradient oracle ∇f, $0 < \mu < L$, integer $N \geq 1$

$\Delta \leftarrow L - \mu$, $\eta \leftarrow (L + \mu)/\Delta$, $u \leftarrow 1/\eta$
$x_1 \leftarrow x_0 - \frac{2}{L+\mu}\nabla f(x_0)$
for $s = 1, \ldots, N-1$ **do**
    $a^2 \leftarrow \dfrac{\sin^2(\pi s/N)}{4\sin(\pi(s - \frac{1}{2})/N)\sin(\pi(s + \frac{1}{2})/N)}$
    $d \leftarrow \eta - a^2 u$, $\beta \leftarrow a^2 u/d$, $\gamma \leftarrow 2/(\Delta d)$
    $x_{s+1} \leftarrow x_s + \beta(x_s - x_{s-1}) - \gamma \nabla f(x_s)$
    $u \leftarrow 1/d$
end for
return x_N